FORECASTING COVID-19 WITH TEMPORAL HIERARCHIES AND ENSEMBLE METHODS

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by

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ABSTRACT

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Infectious disease forecasting efforts underwent rapid growth during the COVID-19 pandemic, providing guidance for pandemic response and about potential future trends. Yet despite their importance, short-term forecasting models often struggled to produce accurate real-time predictions of a complex and rapidly changing system. This gap in accuracy, which persists three years into the pandemic, warrants the exploration and testing of new methods to glean fresh insights.

In this work, we examine the application of the temporal hierarchical forecasting (THieF) methodology to probabilistic forecasts of COVID-19 incident hospital admissions in the United States. THieF is a recent, innovative forecasting technique that aggregates time-series data into a hierarchy made up of different temporal scales, produces forecasts at each level of the hierarchy, then reconciles those forecasts using optimized weighted forecast combination. While THieF's unique approach has shown substantial accuracy improvements in a diverse range of applications, such as operations management and emergency room admission predictions, this technique has not previously been applied to outbreak forecasting.

We generate candidate models formulated using the THieF methodology, which differ by their hierarchy schemes and data transformations, and ensembles of the THieF models, computed as a mean of predictive quantiles. The models are evaluated using weighted interval score (WIS) as a measure of forecast skill, and the top-performing subset is compared to several benchmark models. These models include simple ARIMA and SARIMA models, a naive baseline model, and an ensemble of operational incident hospitalization models from the US COVID-19 Forecast Hub. The THieF models and THieF ensembles demonstrate improvements in WIS and MAE, as well as competitive prediction interval coverage, over the benchmark models for both the validation and testing phases. The best THieF model generally ranked second out of nine total models during the testing evaluation. These accuracy improvements suggest the THieF methodology may serve as a useful addition to the infection disease forecasting toolkit.

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CHAPTER 1

INTRODUCTION

Throughout the past three years, the COVID-19 pandemic has illuminated repercussions of infectious diseases beyond mortality alone. This type of outbreak can strain public health systems, burden economies, and result in other individual and societal costs. During the early years of the pandemic, infectious disease forecasting served as an important bridge between the scientific community, policymakers, and general public to communicate information about potential future directions of the pandemic.

Substantial forecasting efforts for outbreaks of other transmissable diseases—including dengue, ebola, SARS, and influenza—have also been undertaken to reduce their spread and wider impact[1], in which modelers make quantitative predictions about the future based on already observed data. These predictions usually pertain to a specific event, outcome, or trend related to one (though occasionally more) infectious disease[13]. Decision-makers may use infectious disease forecasts to effectively balance prevention or mitigation efforts[18]. For example, forecasted trends may help determine hospital staffing or allocation of limited resources; influence the timing of targeted communication campaigns about social distances or vaccination to predate predicted increases; or inform decisions to ease lockdowns to boost the economy during downward trends.

COVID-19 (and other infectious disease) forecasts are short-term—predicting no more than four weeks into the future—quantifiable, and specific for a fixed period of time. Therefore, these forecasts may be evaluated through comparisons to the eventually observed data[7, 13]. While longer-term COVID-19 scenario projections under specific, hypothetical conditions have great utility in more lasting plans to combat infectious disease, we restrict the scope of this work to short-term COVID-19 forecasts.

Short-term COVID-19 forecasting has been heavily researched in the past three years, with efforts synthesized, collected, archived, and evaluated in a centralized repository called the U.S. COVID-19 Forecast Hub[6] (referred to as the Forecast Hub from here on). COVID-19 forecasts are mainly probabilistic, a distinguishing characteristic of outbreak forecasting efforts. This type of forecast is more informative than a simple point forecast because probabilistic forecasts quantify the likelihood of their occurrence[7, 6, 13].

Producing accurate forecasts, however, is not an easy task. The systems that govern the spread of infectious disease tend to be non-stationary due to inherent factors like exponential growth. Partial dependence on human behavior also contributes to this non-stationarity[13]. For COVID-19, this non-stationarity was a result of continually changing policies and behaviors due to school closures, lock downs, masking, social distances, and the appearance of new variants with different characteristics[7]. Trends of non-stationary processes are more difficult to forecast. Further, COVID-19 forecasts were used to inform such shifting policies, which induced the forecasting feedback loop. The forecasting feedback loop describes a paradox that results when infectious disease forecasts guide intervention or risk communication, which in turn impacts the path of the epidemic being forecast[7, 13].

COVID-19 forecasting also initially faced the challenges inherent for an emergent outbreak setting, in which data availability and quality is severely limited[13]. While the availability and quality of data improved throughout the pandemic, data accuracy concerns still remain due to a lack of national testing protocols and issues with properly categorizing deaths. For these reasons and reporting requirements for Medicare reimbursement, incident hospitalizations became a more relevant public health metric than incident cases and incident deaths, despite fewer initial forecasting efforts for this target. The ceasing of data collection and reporting for COVID-19 incident cases and deaths by Johns Hopkins University (the official ground truth data source) also points to incident hospitalizations as the most appropriate forecast target to investigate [9, 8, 13].

Further, prior COVID-19 forecasting research has shown most models struggle to accurately predict trend fluctuations in which the slope or direction of the data change from the previously held pattern. Even with a range of data sources and modeling techniques, many models from the Forecasts Hub do not assign a probability to the potential of a large change in trend [7]. This gap clearly necessitates the exploration and testing of new methods.

Temporal hierarchical forecasting (THieF) is one such method that uniquely combines hierarchical forecasting and forecast combination for time series data. This approach, developed by Athanasopoulos, et al. in 2017, consists of three main steps: creation of a temporal hierarchy through data aggregation, generation of forecasts at each level of the hierarchy with the classic statistical model, and reconciliation of those initial forecasts. The final step performs weighted forecast combination to not only achieve coherence but also improved accuracy[1]. Coherence describes when hierarchical forecasts predict the same trend, event, or outcome at every level such that the forecasts at lower levels add up to those at higher levels. Reconciliation describes the process of making forecasts coherent. When forecasts from different levels of the hierarchy disagree about the future (are not coherent), it is unclear which level's forecasts should be trusted and chosen to inform decision-making[12].

Based on the challenges that infectious disease forecasting faces, we propose that the THieF methodology may have something to offer. First, THieF relies on a phenomenological model, a class of models that use only observed data to make predictions, which have shown good performance for short-term outbreak forecasting[13]. Second, to our knowledge, THieF has yet to be applied to outbreak forecasting. Hence, this methodology may offer insights about effective modeling techniques. Third, THieF has shown significant performance gains in other settings, which may translate to infectious disease forecasting[7, 1].

We use models made with the THieF methodology and ensembles of the THieF models to retrospectively forecast for COVID-19 daily incident hospitalizations in the United States at both the state and national scale. These original THieF models, the THieF ensembles, and a group of Autoregressive Integrated Moving Average (ARIMA) and season ARIMA (SARIMA) models make up a total of twenty-six (26) candidate models evaluated during the validation phase. A most-accurate subset of up to nine (9) models are passed on to the testing phase where they are compared to a naive baseline and an untrained median ensemble. Success is demonstrated by the THieF ensembles and a few original THieF models that showed substantial accuracy gains over the baseline, with one of the models also displaying several modest improvements over the ensemble.

CHAPTER 2

METHODS

2.1 Surveillance data

To conduct this analysis, we use confirmed hospital admissions from HHS Protect data from the COVID-19 Reported Patient Impact and Hospital Capacity by State Time Series. This dataset reports state-aggregated daily incident hospitalizations starting on January 1, 2020 for U.S. states and territories. The data come from three primary sources: HHSTeleTracking, state/territorial health departments which report to HHS Protect, and National Health Care Safety Network (prior to July 15, 2020) [19].

On December 1, 2020, the U.S. COVID-19 Forecast Hub designated this dataset as the official ground truth data for COVID-19 incident hospitalizations. (No other sources were designated official ground truth data for incident hospitalizations prior to this date) [6]. This meant that the HHS Protect data data was used to evaluate the performance of real-time incident hospitalization forecasts submitted to the Forecast Hub. Hence, teams were encouraged to use this hospitalization data set to train their models [7]. In the analysis, we also use this dataset as both input training data for our models and truth data against which to perform accuracy evaluations.

Hospitalization reporting was sporadic for the first four months of the pandemic, with consistent, reliable values for major states and territories achieved only in late July 2020. Specifically, reports of daily confirmed hospitalizations for all 50 states, Washington D.C., Puerto Rico, and the U.S. as a whole began on July 27, 2020. Although COVID-19 hospitalization reporting stabilized after that for cases and deaths, hospitalization values are often more reliable due to reporting requirements for Medicare reimbursement. COVID-19 cases have been vastly underreported for the entire duration of the pandemic [9], and this problem has worsened with dwindling testing. Further, truth data collection and reporting for COVID-19 incident cases and deaths has been discontinued by Johns Hopkins University (the official ground truth data source) as of March 10, 2023 [8].

Surveillance data, like the HHS Protect data confirmed hospitalizations, sometimes requires revisions due to reporting anomalies. Hospitalizations might be entered on incorrect dates or added after COVID-19 was determined to be the cause of hospitalization. Updates to values usually occurred within the first week or so of posting. As a result, real-time forecast models might train on data that include incident hospitalization numbers that do not reflect the finalized values.

We query these confirmed hospitalizations from the *covidData* package through the *covidHubUtils* package, both developed by Reich Lab at the University of Massachusetts Amherst to deal with COVID-19 surveillance data. Versions of this data are stored such that users may access the truth data as of a specified date[3]. For example, it is possible to retrieve of version of the confirmed hospitalization data set as of July 27, 2020. To ensure fair comparisons between the models developed for this analysis and some real-time models, our models are trained on the confirmed hospitalizations that would have been available as of the "forecast date" of the retrospective forecast.

2.2 Forecasts

A forecast for a date-target-location combination (e.g. "1-day ahead incident hospitalizations in Massachusetts relative to 2020-07-04") is defined by one or both of a point forecast and a probabilistic forecast, represented by a set of 23 quantiles [7]. (More information is given about forecast targets, horizons, locations, and quantiles in

Sections 2.2.1, 2.2.2, and 2.2.3.) According to conventions set by the Forecasts Hub, all forecast values are truncated to be non-negative, as there is no real-world meaning for negative incident hospitalizations [4]. Further, while this is a retrospective analysis, forecasts are made on Sunday weekly (as they would be if made in real-time) using the data available as of that day. We chose to make forecasts on Sunday because confirmed hospitalization values for subsequent days did not update until Tuesday or later until some time in March 2021, which presented several issues. First, if we began validation forecasting after March 2021, we would lose three months of the period of analysis. Second, the main comparison models made forecasts on Monday, meaning that any models that forecast the week before would be at a disadvantage while any models that forecast on Tuesday or later would be evaluated with the following week's forecasts. Third, THieF does not allow for missing data points.

2.2.1 Forecast horizons

Short-term forecasts for COVID-19 have been defined to encapsulate horizons of 1- to 4-weeks ahead, or, equivalently, 1- to 28-days ahead [7]. Forecasts for incident hospitalizations are made on a daily time scale, so we choose to restrict[1] the forecast horizons we evaluate to 1- to 28-days ahead.[2] Horizons are related to target end dates in that a target end date of a particular forecast may be calculated by adding its horizon to the forecast date.

2.2.2 Forecast locations

We make forecasts for 53 of 55 locations from the HHS Protect data confirmed hospitalizations dataset during the period of analysis. These 53 locations include the 50 states, four jurisdictions/territories (Washington D.C. and Puerto Rico), and a U.S.

[1] Some models make up to 84-day ahead forecasts due to the THieF methodology construction.

[2] Note, there is a slight exception to this restriction. See Section 2.6.4 for more information.

national level. American Samoa and Guam are excluded due to their very low or zero COVID-19 incident hospitalizations over the entire pandemic. Forecasts for low-count locations that predicted single-digit or zero hospitalizations were often very accurate; hence, these locations do not offer meaningful contributions to our understanding of differences in forecast skill between modeling approaches [7]. Further, even though we include the U.S. national level, the remaining 52 locations are our primary focus, as the magnitude of national forecasts is much greater than out of any single state.

2.2.3 Forecast quantiles

The Forecast Hub specifies the usage of 23 quantiles

$$\mathbb{Q} = \{.010, 0.025, .050, .100, \cdots, .900, .950, .975, .990\}$$

to represent a full probabilistic distribution, with the median 0.500 quantile usually taken as the model's point forecast. We follow this convention, though some additional work is needed to obtain probabilistic forecasts from the THieF methodology (see Section 2.5.4).

2.3 Date range of analysis

We focused on the period between Monday, July 27, 2020 and Sunday, October 2, 2022. This date range spans 798 days, or 114 weeks, which is just over 26 months. The entire period of analysis is split into two main phases: a validation phase of 462 days (66 weeks) and a testing phase of 236 days (48 weeks). The split between these two phases is such that the validation phase ends on Sunday, October 31, 2021 and the testing phase begins on Monday, November 1, 2021.

The first day of the period of analysis (July 27, 2020) was selected as the first day of reliable truth data for all locations of interest. Forecasting for the validation phase began on Monday, December 7, 2020, as this is the first Monday after the HHS Protect

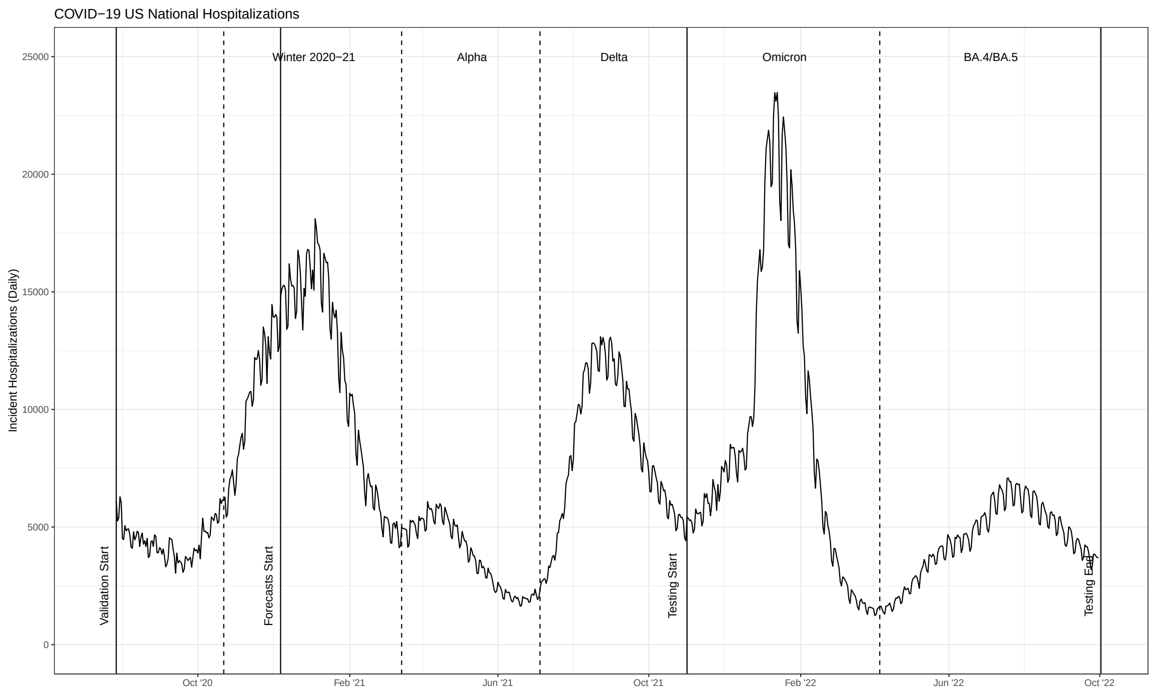

Figure 2.1. A plot of the period of analysis, superimposed over COVID-19 incident hospitalizations for the entire United States. Vertical solid lines indicate divisions between phases while dashed lines indicate divisions between variant waves.

data confirmed hospitalizations data set was declared the official ground truth data. The division between the validation and testing phases was chosen to encapsulate three COVID-19 variant-based waves in the validation phase (Winter 2020-21, Alpha, and Delta) and two waves in the testing phase (Omicron and BA.4/BA.5). The separation between pandemic waves is made based on U.S. national level incident hospitalizations, pictured below in Figure 2.1.

2.4 Benchmark comparison models

Two models from the Forecast Hub—the COVIDhub-baseline and COVIDhub-4_week_ensemble—are selected to serve as comparisons for the models in this analysis during the testing phase evaluation. Many of the metrics used in this evaluation are not especially meaningful on their own, the value often dependent on the magnitude of the observed data, requiring a relative comparison among models forecasting the same targets (1- to 28-day ahead incident hospitalizations for COVID-19) [7].

The COVIDhub-baseline is a naive baseline model created and run by the U.S. COVID-19 Forecast Hub, designed as a simple benchmark for comparison. The baseline makes forecasts using the previous day's incident hospitalization value as the point forecast (and 0.5 quantile for the probabilistic forecast) and using past changes in daily incidence to calibrate uncertainty in its probabilistic forecasts [7]. Beating the baseline is typically seen as one marker of a useful model.

The COVIDhub-4_week_ensemble is an equally-weighted median ensemble of eligible[3] submitting models for incident hospitalizations at the COVID-19 Forecast Hub, run by the Hub team that includes forecasts up to 28-day (4-week) ahead horizons. This model produces the same 1- to 14-day (1- to 2-week) ahead horizon forecasts as the COVIDhub-ensemble model, with the addition of longer horizons [4]. These 15- to 28-day (3- to 4-week) horizons were removed from the main ensemble model, as research showed a sharp decline in accuracy in forecasts predicting beyond 2-week aheads [16]. However, we still include such longer horizons in the analysis for comparison with our models.

Throughout the period of analysis, the COVIDhub-4_week_ensemble consisted of a shifting list of contributing models due to changing eligibility criteria and submitting models for incident hospitalizations [4]. This may have had an impact on shifting accuracy of the ensemble, though the extent is unclear.

The forecasts for the two comparison models are obtained from the Zoltar forecast archive, a research data repository developed by Reich Lab, through the *covidHubUtils* package [20].

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[3]For more information about forecast eligibility, see [4]

2.5 Temporal hierarchical forecasting (THieF)

Historically, hierarchical forecasting has been utilized in business, economics, operations management, and similar fields. This type of forecasting brings advantages like producing coherent forecasts (hierarchical forecasts are coherent if the forecasts at lower levels sum up to those at higher levels)[4] and increasing prediction accuracy compared to other methods [1, 12]. While most hierarchical forecasting considers cross-sectional hierarchies, not temporal ones, many of the same concepts may be applied to both types of hierarchies. Like with hierarchical forecasting, forecast combination has shown favorable outcomes in a myriad of applications[1]. The unique merging of forecast combination with hierarchical forecasting in the THieF methodology has resulted in promising accuracy improvements over more traditional techniques that solely involve hierarchical forecasting [1].

Forecast combination that integrates forecasts from different, independent modeling frameworks to create ensembles often results in better infectious disease forecasts compared to those from individual models [15]. Likewise, classical statistical models like ARIMA models have demonstrated good performance for short-term forecasts in outbreak settings [13]. THieF's built-in forecast combination (which shares some key similarities with that of ensemble methods) and underlying ARIMA base-forecaster suggest that this methodology will show accuracy improvements when forecasting COVID-19 incident hospitalizations.

Surveillance data tends to be quite noisy due to reporting artifacts, especially when considering small time scales. COVID-19 incident hospitalizations suffer a day-of-the-week effect as a result of hospital and clinic operating hours. Fewer hospitalizations are reported on Saturday and Sunday, instead often recorded as part of Monday hospitalizations, since not all facilities admit patients on weekends and those that do

[4]Note that coherence is not an intrinsic property of all hierarchical forecasts.

may not report those hospitalizations until the next week. We may better capture true short-term trends by aggregating daily hospitalizations to be weekly or every two weeks, although such smoothing carries some risk of excluding useful information at the daily level. Thus, we are careful to include both daily and weekly aggregation levels in the THieF temporal hierarchy construction for every model. Other higher aggregation levels are included in model schemes to investigate the existence of hidden patterns at higher time scales and if such a pattern could improve model accuracy [10].

The THieF methodology consists of three steps to produce forecasts from time-series data using a time-based hierarchical structure. These steps—data aggregation and hierarchy construction, base forecast creation, and reconciliation of base forecasts—are outlined in further detail below and in Figure 2.2[1].

2.5.1 Data Aggregation and Hierarchy

Suppose we are interested in constructing a THieF model that combines forecasts from 7-day and 14-day intervals because we believe these time scales each contain important information. Aggregate series for each of these times scales can be constructed from the original daily time series by summing the observed values over a particular time scale[1]. All of our models define the lowest (and least aggregate) level to be the original daily time scale, so our example model has three total levels: 1-day, 7-day, 14-day (see Figure 2.2). Although each level in this example is a multiple of all the levels below it, any k-day time scale can be selected for the hierarchy as long as k is a factor of 14. For example, a 2-day aggregation would also be allowed in this temporal hierarchy. On the other hand, 3- or 5-day aggregations cannot be added to the existing levels since these values are not factors of the topmost 14-day aggregation. The reason for this rule is that non-factor levels result in non-integer seasonality and thus impede computation[1].

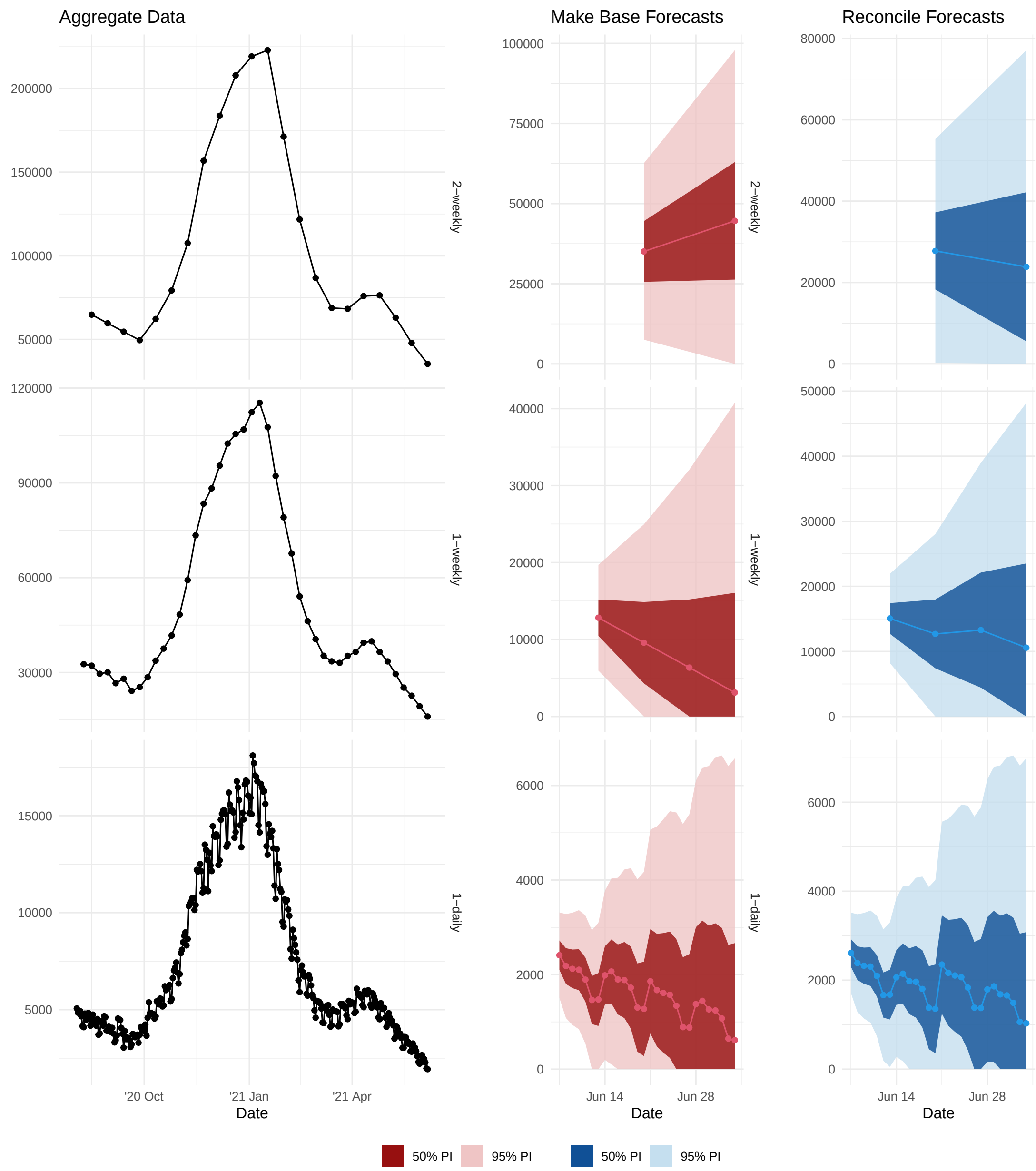

Figure 2.2. A plot showing the three steps of the THieF methodology for forecasting U.S. national incident hospitalizations for 1 to 28 days ahead on June 7, 2021

The hospitalization data for a particular location may be expressed generally as a time series $\{y_t; t = 1, \cdots, T\}$ when observed at the daily level. Let m be the number of observations of the smallest timescale (and the highest frequency) that aggregate to create a single data point for the the most-aggregate level. (In our example $m = 14$.) These data are then aggregated to different k-day time scales based on the aggregation levels of interest for a particular model (e.g. 7-day, 14-day, 28-day, 42-day, etc.[5])[1]. A data point with index j for a k-day aggregated series $y_j^{[k]}$ may be obtained using

$$y_j^{[k]} = \Sigma_{t=t^*+(j-1)k}^{t^*+jk-1} y_t$$

for $j = 1, \cdots, \lfloor T/k \rfloor$ and seasonal period $M_k = m/k$ for that series. Here we can see that the variable $k \in \{k_p, \cdot, k_2, k_1\}$ can take on multiple values and indicates the particular aggregation level, where $k_p = m$, $k_1 = 1$, and p is the total number of levels in the hierarchy[1]. In our example from earlier, $k \in \{14, 7, 1\}$, where $p = 3$.

The use of $t^* = T - \lfloor T/m \rfloor m + 1$ ensures even, non-overlapping aggregation where the total number of observations is a multiple of m. In practice, this means that the remainder $r = t^* - 1$ from dividing T by m is the number of observations that are removed from the beginning of the original time series before aggregation occurs. For example, if we start with $T = 31$ observations in a daily time series and are interested in $k \in \{14, 7, 1\}$-day aggregation levels, we remove the first $t^* - 1 = (31 - \lfloor 31/14 \rfloor 14 + 1) - 1 = 4 - 1 = 3$ data points. We may then begin aggregation and can calculate the first data point for the 7-day time series with the equation $y_1^{[7]} = \Sigma_{t=4+(0)7}^{4+7-1} y_t = y_4 + \cdots + y_{10}$, which is as we expect given that the first $t^* - 1 = 3$ observations are removed.

Notice that the index j used above is aggregation-level specific, with no easy way to translate between levels of the hierarchy. Hence, Athanasopoulos, et al. propose

[5] see Section 2.6.1 for a full list of time scales considered

an alternative index i where i is the original observation index of the top-level, most aggregate series $y_i^{[m]}$ for $i = 1, \cdots, \lfloor T/m \rfloor$ so that $i = j$. A more general form for any level k becomes $y_{M_k(i-1)+z}^{[k]}$ for $z = 1, \cdots, M_k$. An increase in the index i by 1 causes an increase of M_k periods for the time series at all aggregation levels, while an increase in z denotes a single period increase for the k-day aggregation level.

Once again using our example, we can show that the new notation is consistent with the old notation. The first observation ($z = 1$) at the $k = 7$-day level of the second 14-day period ($i = 2$) can be written in the new notation as $y_{M_7(2-1)+1}^{[7]} = y_{2(2-1)+1}^{[7]}$. This can be shown to simplify down to $y_{2+1}^{[7]} = y_3^{[7]}$, which is exactly how this observation (the third overall observation in the 7-day series) is written in the original notation for $j = 3$.

For every index i, we construct $p - 1$ column vectors of dimension $(M_k \times 1)$

$$\boldsymbol{y_i}^{[k]} = (y_{M_k(i-1)+1}^{[k]}, y_{M_k(i-1)+2}^{[k]}, \cdots, y_{M_k i}^{[k]})',$$

for all but the topmost level which only has a single observation for each i.

These column vectors are then stacked into a larger $(\Sigma_k M_k \times 1)$ column vector $\boldsymbol{y_i} = (y_i^{[m]}, \cdots, \boldsymbol{y_i}^{[k_3]'}, \boldsymbol{y_i}^{[k_2]'}, \boldsymbol{y_i}^{[1]'})'$ such that

$$\boldsymbol{y_i} = \boldsymbol{S}\boldsymbol{y_i}^{[1]}$$

where $\boldsymbol{S}$ is the $(\Sigma_k M_k \times m)$ summing matrix from [11] and $\boldsymbol{y_i}^{[1]}$ is the $(m \times 1)$ vector of daily observations with index i. For a hierarchy with levels $k \in$ 14, 7, 1, this equation can be expanded out to be

$$\begin{bmatrix} y_i^{[14]} \\ \boldsymbol{y_i}^{[7]} \\ \boldsymbol{y_i}^{[1]} \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 \\ & & & & I_{14} & & & \end{bmatrix} \begin{bmatrix} \boldsymbol{y_{14(i-1)+1}^{[1]}} \\ \boldsymbol{y_{14(i-1)+2}^{[1]}} \\ \cdots \\ \boldsymbol{y_{14i}^{[1]}} \end{bmatrix}$$

in which the dimensions of the matrices are (17×1), (17×14), and (14×1), respectively.

2.5.2 Base Forecasts

Before we can obtain coherent forecasts, we must begin by making base forecasts at every level of the hierarchy, as shown in Figure 2.2. Each level's base forecasts are independent of the other levels and are created as if each aggregation level is a separate, univariate time series. Theoretically, this means that different types of models could be used to make forecasts at every level prior to reconciliation, although in practice we generally use the same model type but allow the values of the parameters to vary between levels. Since the parameter values may vary, this usually results in incoherent forecasts at this stage. Such disagreement between the forecasts may indicate that important patterns have been detected at different time scales, suggesting the THieF method may be useful.

No particular forecasting framework is specified in the original equations for the THieF methodology, but the authors later show that an underlying ARIMA model produces the most accurate forecasts. An ARIMA model is also used in the implementation of THieF in the associated R package[1]. Hence, we also choose to use ARIMA models to produce our base forecasts.

Although we are only interested in evaluating up to 28-day ahead forecasts from our models, we must make forecasts at every level with horizons at least as large as the top, k_p-day aggregation to make use of the THieF methodology. We use g^* to represent our maximum required forecast horizon in terms of days. Then we construct an index g based on the top-level series, just as we did with the index i for the observed series in the previous section, such that $g = 1, \cdots, \lceil g^*/m \rceil$[1]. Based on the $\lfloor T/k \rfloor$ observations for $z = 1, \cdots, M_k$, we make $\lceil g^*/k \rceil$ forecasts for each k-day aggregation level of the form $\hat{y}_{M_k(g-1)+z}$. Like with the observed data, we then construct $(\Sigma_k M_k \times 1)$ column vectors for each forecast horizon g:

$$\hat{\boldsymbol{y}}_{\boldsymbol{g}} = (\hat{y}_g^{[m]}, \cdots, \hat{\boldsymbol{y}}_{\boldsymbol{g}}^{[k_3]\prime}, \hat{\boldsymbol{y}}_{\boldsymbol{g}}^{[k_2]\prime}, \hat{\boldsymbol{y}}_{\boldsymbol{g}}^{[1]\prime})'$$

where each element $\hat{\boldsymbol{y}}_{\boldsymbol{g}}^{[k]} = (y^{[k]}_{M_k(g-1)+1}, y^{[k]}_{M_k(g-1)+2}, \cdots, y^{[k]}_{M_k g})'$ is a $(M_k \times 1)$ column vector of the base forecasts from that level [1].

The base forecasts may also be expressed as

$$\hat{\boldsymbol{y}}_{\boldsymbol{g}} = \boldsymbol{S}\boldsymbol{\beta}(g) + \boldsymbol{\varepsilon}_{\boldsymbol{g}}$$

where $\boldsymbol{\beta}(g) = \boldsymbol{E}[\boldsymbol{y}^{[1]}_{\lfloor T/m \rfloor + g} | y_1, \cdots, y_T]$ is the unknown mean of future daily observed values conditional on the currently observed data points; and $\boldsymbol{\varepsilon}_{\boldsymbol{g}}$ is the difference between the base forecasts $\hat{\boldsymbol{y}}_g$ and the expected value of their corresponding reconciled forecasts, termed the reconciliation error, assuming that ε_g has zero mean and covariance matrix $\boldsymbol{\Sigma}$ [1].

2.5.3 Reconciled Forecasts

Through the step of reconciliation, we integrate the information from all of the aggregation levels into the output forecasts. Ideally, we want to combine the base forecasts $\hat{\boldsymbol{y}}_{\boldsymbol{g}}$ in a way that minimizes the difference between them and the reconciled forecasts $\tilde{\boldsymbol{y}}_{\boldsymbol{g}}$. If we knew the covariance matrix of the reconciliation error $\boldsymbol{\Sigma}$, we could use a generalized least squares (GLS) estimator to solve this minimization problem; however, $\boldsymbol{\Sigma}$ has been shown to not be identifiable[1]. Instead, we substitute a weighted least squares (WLS) estimator $\boldsymbol{\Lambda}_{\boldsymbol{SV}}$ called the series variance estimator to obtain the reconciled forecasts

$$\tilde{\boldsymbol{y}}_{\boldsymbol{g}} = \boldsymbol{S}\hat{\boldsymbol{\beta}}(g) = \boldsymbol{S}(\boldsymbol{S'}\boldsymbol{\Lambda}^{-1}_{SV}\boldsymbol{S})^{-1}\boldsymbol{S}\boldsymbol{\Lambda}^{-1}_{SV}\hat{\boldsymbol{y}}_{\boldsymbol{g}},$$

where $\boldsymbol{S}$ is the previously defined summing matrix and $\hat{\boldsymbol{\beta}}(g)$ is an estimator of the unknown conditional mean $\boldsymbol{\beta}(g)$[1].

The series variance estimator is a diagonal $(\Sigma_k M_k \times \Sigma_k M_k)$ matrix of weights proportional to the error variance of each aggregated series that make up the levels of the hierarchy. In other words, the series variance estimator estimator assumes that variance is constant across the forecasts for each time-series, a common assumption for time-series data[1]. For the THieF model with the top-most 14-day aggregation level, this matrix is given by

$$\boldsymbol{\Lambda}_{SV} = \mathrm{diag}(\hat{\sigma}^{[14]}, \hat{\sigma}^{[7]}, \hat{\sigma}^{[7]}, \hat{\sigma}^{[1]}, \hat{\sigma}^{[1]}, \hat{\sigma}^{[1]}, \hat{\sigma}^{[1]}, \hat{\sigma}^{[1]}, \hat{\sigma}^{[1]}, \hat{\sigma}^{[1]})^2.$$

While Athanasopoulos, et al. propose two other alternative WLS estimators, they later show that $\boldsymbol{\Lambda}_{SV}$ yields the most accurate reconciled forecasts. All of these WLS estimators are estimators for the sample covariance estimator $\boldsymbol{\Lambda}$ for $\boldsymbol{W}$, the covariance matrix of the base forecast error. In turn, $\boldsymbol{W}$ is an estimator for the unknowable $\boldsymbol{\Sigma}$[1]. The reason we don't use $\boldsymbol{\Lambda}$ is that its calculation is difficult to implement in practice[6]. This estimator has $\Sigma_k M_k^2$ elements that must be estimated, and the sample size that can be used for such estimation is limited by the number of observations at the topmost aggregation level $\lfloor T/m \rfloor$. Since $\lfloor T/m \rfloor \ll T$, the accuracy of the estimation is degraded. Hence, we instead choose to rely on $\boldsymbol{\Lambda}_{SV}$[1].

Following this final step of reconciliation, the forecasts from every time scale will agree about the number of daily incident hospitalizations for all horizons. The reconciled forecasts are shown in Figure 2.2.

[6]This estimator also cannot be calculated for non-model based forecasts like expert judgements in which in-sample forecast errors may not be available[1], though this is not a concern for our work.

2.5.4 Making THieF Probabilistic

Recall that a defining feature of COVID-19 forecasts is that they are probabilistic. However, the THieF defined above and in [1] is used for point forecasting. To address this challenge, we calculated prediction intervals for the base forecasts, reconciled the the interval end points using the same weighted least squares procedure described in the previous section, then translated the prediction interval bounds into quantiles from a full probabilistic distribution. The translation is summarized in Table 2.1. For example, the values for a 0.500 and a 0.950 quantile (see the Quantile row) are obtained by taking the lower and upper bounds of the 95% prediction interval (see the Interval row, where "L" and "U" represent the lower and upper bounds, respectively).

Although it would be preferable to obtain quantiles from the reconciled forecasts themselves, we generally did not find a concerning drop in prediction interval coverage rates for our evaluations conducted during either the validation phase or the testing phase (see Section 3).

Quantile	.010	.025	.050	0.100	...	0.500	...	0.900	0.950	.975	.990
Interval	98L	95L	90L	80L	...	Median	...	80U	90U	95U	98U

Table 2.1. Table of the 23 quantiles required by the Forecast Hub and their associated prediction interval bound.

2.6 Model specifications

We create three distinct classes of models for comparison during the validation phase: the original THieF models, the THieF ensembles, and the SARIMA models. In total, these three groups consist of twenty-six (26) candidate models, all of which are implemented using the *forecast* package in R.

2.6.1 Original THieF models

The original THieF models are constructed using the THieF methodology outlined in Section 2.5 with different combinations of aggregation levels and data transformations. Seven hierarchical structures and two data transformations result in fourteen (14) original THieF models. Each model is defined (and named) based on the highest level of its temporal hierarchy and the data transformation that is used before generating the forecasts. For example, the simplest of the THieF models is THieF_1wk-noTransform while the most complex is THieF_12wk-4root.

The aggregation levels of any model's temporal hierarchy are daily, plus f_m-week periods where f_m is the set of all factors for the highest time scale m. Hence, the THieF_12wk-4root model (as well as the THieF_12wk-noTransform model) has a temporal hierarchy made up of daily, weekly, 2-weekly, 3-weekly, 4-weekly, 6-weekly, and 12-weekly aggregation levels since $f_{12} = \{1, 2, 3, 4, 6, 12\}$. Meanwhile, the THieF_1wk-noTransform model's temporal hierarchy is only made up of daily and weekly aggregation levels ($f_1 = \{1\}$). We chose 1-week, 2-week, 3-week, 4-week, 6-week, 8-week, and 12-week top-level aggregations so as to have a range of time scales to inform the reconciled forecasts. Supplemental Figure 1 illustrates the construction of several models' temporal hierarchies.

We explore the seven hierarchy schemes with both no data transformation and a fourth-root data transformation. The latter transform is implemented by first taking the fourth root of the confirmed hospitalizations time series, using the truth data to make forecasts with the THieF methodology, then performing an inverse fourth-power transformation to the resulting forecasts. We selected the fourth-root transformation based on prior research that had shown that had shown accuracy improvements for COVID-19 forecasting due to its variance-stabilizing effect [17].

2.6.2 THieF ensemble models

The THieF ensemble models[7] are seven mean ensembles of the fourteen total original THieF models. While not all fourteen original THieF models are passed onto the testing phase, the THieF ensembles that are passed on remain ensembles of these fourteen original THieF models (see Section 2.6.6).

Of the seven THieF ensembles, we formulated six as trained ensembles, in which past performance of the component THieF models is used to assign the components weights, and one as an untrained ensemble, in which all component forecasters are assigned equal weight [14]. Both types of ensembles make quantile forecasts[8] at level u by combining those of the component forecasts for a particular forecast date, location, horizon, and quantile combination. (For example, a particular model d might make a 0.500 quantile forecast on December 7, 2020 that predicts the number of 1-day ahead incident hospitalizations in Massachusetts will be 185.) An ensemble quantile forecast can be expressed mathematically as

$$q^{ens}_{l,s,h,u} = f(q^1_{l,s,h,u}, \cdots, q^D_{l,s,h,u}),$$

where l is the location, s is the forecast date, h is the horizon, and the superscript $d = 1, \cdots, D$ denotes the model [14].

A trained, weighted mean ensemble calculates its forecast quantiles as

$$q^{ens}_{l,s,h,u} = \Sigma^D_{d=1} w^d_s q^d_{l,s,h,u},$$

in which the weights are calculated using a sigmoidal transformation of the component models' relative WIS (rWIS) over a rolling window of twelve (12) weeks leading up

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[7]The equations used to define the ensembles are adapted from Ray, et al. [14]

[8]Recall that we use the 0.500 quantile forecast as the point forecast.

to the ensemble's forecast date s:

$$w_s^d = \frac{exp(-\theta \cdot rWIS_s^d)}{\Sigma_{d'=1}^{D} exp(-\theta) \cdot rWIS_s^{d'}}.$$

When the non-negative parameter $\theta = 0$, this becomes an equally weighted ensemble. However, as θ increases, the weights become more unequal, with the contributions of better-performing component models eclipsing that of worse-performing models. Note that the weights, however, are always non-negative and sum to one. A group of weights is also calculated for a particular forecast date, location, horizon, and quantile combination[14].

The THieF ensembles are named with the θ value used in their construction, except for the untrained ensemble, which is called THieF_ensemble-mean. We select θ values {0, 1, 3, 6.5, 10, 15, 20, 25} to encompass the range of values taken on by the COVIDhub-trained_ensemble[9] when predicting incident hospitalization [14]. The one non-integer value (6.5) was selected as the median (and approximate mean) θ value over the period of analysis for all locations of interest except for the U.S. national level. The corresponding ensemble was named THieF_ensemble-train6.5.

While the value of θ generally remains static for each ensemble, all trained ensembles are defined to start with $\theta = 0$ until 12 weeks of prior forecasts are achieved (on March 1, 2021) during the validation phase that can be used to calculate weights based on prior performance of forecasts. During the testing phase, the value of the weights is dependent on at least some validation phase performance until the thirteenth week (beginning on January 24, 2022).

[9] A trained ensemble created by the Forecast Hub that makes short-term forecasts for incident cases, incident and cumulative deaths, and incident hospitalizations. This model makes its forecasts by computing a weighted median of the ten best component forecasts (by WIS) in the 12 weeks leading up to the forecast date. For this model, θ is allowed to vary by forecast date [4, 14].

2.6.3 ARIMA and SARIMA models

The four ARIMA and SARIMA (seasonal ARIMA) models are classic statistical models created using the *auto.arima()* function from the *forecast* package in R. This function makes forecasts based on input time series data, and the user may specify seasonality when creating a SARIMA model. Using the original HealthData.Gov confirmed hospitalizations data set, we create two ARIMA models and two SARIMA models, with either no data transformation or a variance-stabilizing fourth root transformation, as described in THieF Models. The SARIMA models both have a seasonality of 7 days (1 week), named sarima_s7-4root and sarima_s7-noTransform, while the ARIMA models have no seasonality, named sarima_s1-4root and sarima_s1-4root.

The construction of the ARIMA and SARIMA models intentionally matches that of the daily and weekly aggregation levels of the original THieF models, though in practice the resulting base forecasts from the THieF models may differ slightly due to the aggregation process to create the temporal hierarchy. We specifically want to use the ARIMA and SARIMA models to help investigate whether it is the THieF methodology's temporal hierarchy and forecast reconciliation that impacts its performance, compared to simply the underlying ARIMA base forecaster. (Recall that research has shown that classic statistical models have shown good performance for outbreak forecasting [13].)

2.6.4 Metrics and Evaluation

Evaluation of our models is conducted by aggregating the scores for the forecasts in seven ways: by forecast date, by location, by pandemic wave, overall validation/testing phase, and in the last 17 weeks (of the validation phase). While we score the forecasts at a daily level, these scores are aggregated by week for most evaluations by the Forecast Hub, and we follow this convention. For each metric, a single value is obtained by averaging the values across a single week (Tuesday through Monday)

[5]. However, modeling teams have made forecasts on different days (e.g. Saturday vs Sunday vs Monday), leading to a mismatch between what day an n-day ahead forecast is predicting for. Hence, for the models we construct, we actually use 3- to 30-day ahead horizons to ensure all forecasts are predicting for the same dates. It should be noted, however, this may result in a slight disadvantage for models whose 3- to 30-day ahead forecasts are being compared to 1- to 28-day ahead forecasts. For example, the ensemble includes some models for which this is the case.

We use the same metrics as the Forecast Hub to evaluate model accuracy for both the validation and testing phases. Average mean absolute error (MAE) is used to score point forecasts, while average weighted interval score (WIS, taken from [2]), average 50% prediction interval (PI), and average 95% PI coverage are used to score probabilistic forecasts. (We refer to these metrics simply as MAE, WIS, and 50% and 95% PI coverage going forward for simplicity.) For WIS and MAE, lower values are desirable while coverage levels approximating $1-\alpha$ indicate a well-calibrated forecast. This means that the intervals are narrow enough to provide useful information but not so narrow as to exclude the truth.

Averages for the four metrics are first calculated for probabilistic forecasts (and point forecasts) for a single forecast date, location, horizon, and quantile combination, then the mean is taken for each model from forecasts aggregated in one of the seven ways mentioned above. These calculations are performed using the following equations, which are adapted from [7]:

1. $MAE = \frac{1}{N}\Sigma_{i=1}^{N}|y_i - \hat{y}_i|$

2. $WIS_{\alpha_{0:V}}(F, y_i) = \frac{1}{N}\Sigma_{i=1}^{N}\frac{1}{V+1/2}(w_0 \cdot |y_i - m| + \Sigma_{v=1}^{V} w_v \cdot IS_{\alpha_v}(F, y_i))$

3. $(1-\alpha) * 100\%$ PI Coverage $= \frac{1}{N}\Sigma_{i=1}^{N}1(l_i^{\alpha} \leq y_i \leq u_i^{\alpha})$,

where N is the number of forecasts being averaged, y_i is the true observed value at a particular time point (date) and location, and $\hat{y}_i$ is the point forecast value for a particular forecast date, horizon, and location combination. The PI uncertainty level is given by α, v is the particular prediction interval ($v = 1, \cdots, V$), and F is the forecast distribution. The weights w_0 and w_v are set as $w_0 = 1/2$ and $w_v = \alpha_v/2$ respectively while u_i^α and l_i^α are the upper and lower bounds of a $(1-\alpha) * 100$-level prediction interval for a particular forecast date and location. Lastly, the interval score of a particular $(1-\alpha) * 100$-level prediction interval is $IS_{\alpha_v}(F, y_i) = (u_i - l_i) + \frac{2}{\alpha} \cdot (l_i - y_i) \cdot 1(y_i < l_i) + \frac{2}{\alpha} \cdot (y_i - u_i) \cdot 1(y_i > u_i)$ where $1()$ is the indicator function[7].

Weighted interval score is a proper scoring rule for a set of interval forecasts comprised of three components—spread, over prediction, and under prediction—that-measures how closely a set of prediction intervals is consistent with the true observed value. WIS is an alternative to more common proper scoring rules like the logarithmic score (logS) and continuous ranked probability score (CRPS) which can not be evaluated directly for interval forecasts. However, with the weights $w_0, \cdots, w_V$ set as described above, large V, and equally spaced values of $\alpha_1, \cdots, \alpha_V$, it can be shown that $\text{WIS}_{\alpha_{0:V}}(F, y_i) \approx \text{CRPS}(F, y_i)$ [2].

Relative versions of MAE and WIS are also used to adjust for truth data magnitude to allow for comparison between forecast values of different scales. These relative metrics are calculated as $MAE_{\text{model d}}/MAE_{\text{baseline}}$ and $WIS_{\text{model d}}/WIS_{\text{baseline}}$, respectively [7].

Evaluation for each of the seven aggregation categories (except for individual locations) are separated by geographic scale: U.S. national and averaged states. The truth data and resulting forecasts are of different magnitudes for the aggregated U.S. national and average states levels. This means that the contribution from the U.S. national location would account for an overwhelming proportion of the resulting forecast scores if we did not make such a split.

We perform no statistical tests to assess the significance of performance differences between models, as certain issues arise when making time series forecasts that prevent usage of such tests. Namely, scores tend to be highly-correlated across multiple dimensions, like horizons, forecast dates, locations, etc. Assumptions of independent, identically distributed data found in standard inference are invalid in time series forecasting. In practice, numerical comparisons of forecast error between prospective models (without excluding any models that were compared) is sufficient for choosing the best model to inform decision-making, and foregoing formal statistical tests is common practice in infectious disease forecasting. Still, it should be noted that we cannot necessarily make a definitive statement about the predominance of one model over another without formal statistical testing [13].

2.6.5 Validation phase

The validation phase serves as a period during which to train the candidate models, calibrate the forecasts, and to select the best models to pass on to the testing phase based on performance during the validation phase. All validation phase evaluations are restricted to forecasts with target end dates (date for which the prediction of incident hospitalizations is being made) within the validation phase (July 27, 2020 to October 31, 2021), with forecasts beginning on Monday, December 7, 2020.

We pre-specify a set of criteria that help us investigate several objectives of interest to help us determine which models to pass along as part of the best subset:

- **Best (THieF) model overall:** Best original THieF over the entire validation phase (for each geographic scale)
- **Does ensembling THieF improve performance:** Best trained THieF ensemble over the entire validation phase (for each geographic scale)
- **Does THieF improve upon performance compared to its base forecaster:** Best ARIMA/SARIMA over the entire validation phase (for each geographic scale)

- **Do some THieF models improve with more input data:** Best original THieF during the last 17 weeks of the validation phase (for each geographic scale)

- **Can weighted ensembles improve upon performance over unweighted ones:** Untrained THieF ensemble

where "best" is defined by lowest WIS. Based on these criteria, a maximum of nine models will make up the best subset to move onto the training phase evaluations, if unique candidates occupy each category. However, if the same model to occupies multiple categories, we may pass on as few as five models.

We stratify by location category due to a difference in magnitude, as described above. For each location category, one model is passed on for each of the model classes—original THieF, THieF (trained) ensemble, ARIMA/SARIMA—as the single best candidate during the entire validation phase. We also select the single best original THieF model during the last 17 weeks of the validation phase, a category designed to account for any models whose performance improves with larger amounts of data on which to train on (since we allow models to use validation phase data to inform their forecasts). This category is limited to THieF models only, as it is models with large aggregation levels that we suspect may be fit this description, and the ARIMA/SARIMA models include only very short seasonality (if any) while the THieF ensemble models incorporate forecasts all the THieF models. The THieF untrained ensemble is passed on regardless of performance, as we are interested in comparing the performance of this model to that of the best trained THieF ensemble.

2.6.6 Testing phase

Using the subset of best candidate models found during the validation phase, we make new testing phase forecasts and evaluate these models against both each other and the two comparison models from the Forecast Hub, COVIDhub-baseline and COVIDhub-4_week_ensemble. Like with the validation phase, all testing phase evaluations are restricted to forecasts with target end dates within the testing phase (November 1, 2021 to October 2, 2022).

The testing phase analysis is broken down into five smaller evaluations, based on aggregating the testing phase forecasts for the overall period, by horizon week, by pandemic wave, by location, and by forecast date, each split by geographic scale (except for by location, where such a split would be redundant). The smaller evaluations are intended to help pinpoint any potential accuracy gains (or losses) the original THieF or THieF ensembles have over the COVIDhub-baseline (or COVIDhub-4_week_ensemble) and allow us to hypothesize the origin of these accuracy differences.

The overall period evaluation provides a generalized view of the testing phase analysis and is meant to identify a single best model that could be used to inform decision-making. (Though of course we cannot say this definitively given that we do not perform any formal statistical tests [13].) The by-horizon evaluation allows identification of models that perform well at shorter horizons (typically 1 to 2 week ahead), as the larger WIS and MAE values for longer horizons (3 to 4 week ahead) will dominate these metrics in the overall period evaluation. This dominance is particularly undesirable given that the Forecast Hub ensemble models, which tend to be some of the most accurate models overall, experience a sharp decline in accuracy beyond two week ahead horizons for incident hospitalizations [16].

The by–pandemic wave and by–forecast date evaluations allow investigation into time periods in which the number of incident hospitalizations is of different magnitudes and is changing at different rates. Times of greater values or most rapid changes have larger WIS and MAE and are generally harder to predict [9, 16], and it may be useful to identify which models perform best under different scenarios. These two evaluations are also stratified by horizon week. However, the by-location evaluation, which provides insight into differences in model performance between higher and lower count locations, is not stratified by horizon week.

CHAPTER 3

RESULTS

3.1 Validation phase

Tables 3.1 and 3.2 show the overall model ranking and evaluation metrics during the validation phase for the averaged states and U.S. national geographic scales, in order of ascending WIS. We can see that for the averaged location scale the THieF ensembles generally perform the best, followed by the original THieF models, then the ARIMA/SARIMA models. This pattern is not as strictly followed for the U.S. national location scale, in which the two original THieF models with a 6-week top aggregation level have lower WIS than at least half of the THieF ensembles.

For averaged states, we pass on THieF_6wk-4root, THieF_ensemble-train3, and sarima_s7-noTransform as the best models of each class during the validation phase overall. While THieF_ensemble-train3 is not strictly the highest ranked THieF ensemble for this location scale, we pass it along based on the selection rule from cross-validation in which the simplest model should be chosen if it does not perform substantially worse than more complex models. For U.S. national, we pass along THieF_6wk-noTransform, THieF_ensemble-train3, and sarima_s1-noTransform as the best representative of each class for the overall validation phase of evaluation. The best original THieF models during the last 17 weeks of the validation phase are THieF_6wk-4root and THieF_12wk-4root for averaged states and U.S. national, respectively, with the full evaluation results shown in Supplemental Tables 2 and 4. The untrained THieF ensemble, while not a top performer for either location scale in the overall analysis, is also passed on to serve as a reference model.

Hence, the best subset that passes onto the testing phase consists of the following seven (7) models: THieF_6wk-4root, THieF_6wk-noTransform, THieF_ensemble-train3, sarima_s7-noTransform, sarima_s1-noTransform, THieF_12wk-4root, and THieF_ensemble-mean.

3.2 Testing phase

The seven models from the best subset are joined by the two Forecast Hub comparison models, the COVIDhub-baseline and the COVIDhub-4_week_ensemble, for the testing phase analysis. Generally, we find that the original THieF models with a fourth root transformation and the COVIDHub-4_week_ensemble show the best performance out of the nine total models across all of the smaller evaluations. The THieF ensembles show more middling performance, though they almost always beat the COVIDhub-baseline.

We illustrate the forecasts made during the testing phase through plotting 1- to 28-day ahead incident hospitalization point, 50% interval, and 95% interval forecasts for the U.S. national location Figure 3.1 from four models of interest: the COVIDhub-4_week_ensemble, the COVIDhub-baseline, a poor performing model (sarima_s7-noTransform), and a top performing model (THieF_6wk-4root). From this figure, we can see that the COVIDhub-4_week_ensemble appears the best follow the true confirmed hospitalizations compared to the other three models, especially during the Omicron wave's period most rapid increase. Note also that for most forecast weeks the COVIDhub-baseline has the widest 95% interval forecasts of any of the displayed models (recall that narrower intervals tend to be more desirable, as long as they capture the true value). The results shown by Figure 3.1 are mirrored (and perhaps further explained) by the following stratified evaluations.

	Model	WIS	MAE	Cov50	Cov95	rWIS	rMAE
1	THieF_ensemble-train10	30.73	47.22	0.53	0.93	0.86	0.94
2	THieF_ensemble-train15	30.74	47.28	0.52	0.93	0.86	0.95
3	THieF_ensemble-train6.5	30.77	47.20	0.53	0.94	0.86	0.94
4	THieF_ensemble-train20	30.77	47.33	0.52	0.93	0.86	0.95
5	THieF_ensemble-train25	30.81	47.37	0.51	0.93	0.86	0.95
6	THieF_ensemble-train3	30.90	47.25	0.54	0.94	0.86	0.95
7	THieF_ensemble-train1	31.33	47.44	0.53	0.94	0.87	0.95
8	THieF_6wk-4root	31.65	48.11	0.46	0.91	0.88	0.96
9	THieF_6wk-noTransform	32.26	48.67	0.54	0.92	0.90	0.97
10	THieF_1wk-4root	32.68	50.06	0.48	0.91	0.91	1.00
11	THieF_2wk-4root	32.94	50.26	0.47	0.91	0.92	1.01
12	THieF_2wk-noTransform	33.26	48.03	0.57	0.90	0.93	0.96
13	THieF_1wk-noTransform	33.74	49.21	0.58	0.90	0.94	0.98
14	THieF_8wk-4root	33.84	51.14	0.43	0.89	0.94	1.02
15	THieF_4wk-4root	34.10	51.50	0.46	0.90	0.95	1.03
16	THieF_ensemble-mean	34.29	47.65	0.53	0.94	0.96	0.95
17	THieF_3wk-4root	34.62	52.71	0.46	0.90	0.96	1.05
18	THieF_8wk-noTransform	34.76	52.06	0.51	0.90	0.97	1.04
19	THieF_3wk-noTransform	34.96	52.08	0.54	0.90	0.97	1.04
20	THieF_4wk-noTransform	35.63	52.82	0.53	0.89	0.99	1.06
21	COVIDhub-baseline	35.88	49.98	0.83	0.98	1.00	1.00
22	sarima_s7-noTransform	36.29	51.91	0.57	0.89	1.01	1.04
23	sarima_s7-4root	36.35	54.55	0.46	0.90	1.01	1.09
24	sarima_s1-4root	36.91	55.43	0.46	0.89	1.03	1.11
25	THieF_12wk-noTransform	39.14	59.06	0.42	0.89	1.09	1.18
26	sarima_s1-noTransform	39.41	56.38	0.54	0.88	1.10	1.13
27	THieF_12wk-4root	84.32	54.49	0.41	0.86	2.35	1.09

Table 3.1. Summary of overall model performance during the validation phase for averaged states, ordered by ascending WIS

	Model	WIS	MAE	Cov50	Cov95	rWIS	rMAE
1	THieF_6wk-noTransform	1116.4	1756.0	0.36	0.91	0.88	0.88
2	THieF_ensemble-train3	1139.8	1808.6	0.34	0.91	0.89	0.91
3	THieF_ensemble-train1	1144.9	1814.9	0.35	0.90	0.90	0.91
4	THieF_ensemble-train6.5	1146.6	1817.4	0.34	0.91	0.90	0.91
5	THieF_6wk-4root	1152.4	1765.8	0.33	0.76	0.90	0.88
6	THieF_ensemble-train10	1159.2	1834.0	0.33	0.91	0.91	0.92
7	THieF_ensemble-train15	1180.2	1866.3	0.33	0.90	0.92	0.94
8	THieF_ensemble-train20	1199.4	1897.5	0.32	0.90	0.94	0.95
9	THieF_ensemble-mean	1212.7	1822.3	0.35	0.90	0.95	0.91
10	THieF_ensemble-train25	1214.3	1919.3	0.32	0.89	0.95	0.96
11	THieF_1wk-noTransform	1218.5	1856.4	0.46	0.87	0.95	0.93
12	THieF_2wk-4root	1234.7	1941.8	0.25	0.85	0.97	0.97
13	THieF_1wk-4root	1252.2	1993.1	0.30	0.85	0.98	1.00
14	COVIDhub-baseline	1276.5	1995.9	0.64	0.99	1.00	1.00
15	THieF_8wk-noTransform	1277.0	1968.9	0.38	0.85	1.00	0.99
16	THieF_8wk-4root	1303.6	1963.8	0.26	0.70	1.02	0.98
17	THieF_2wk-noTransform	1343.1	1985.2	0.40	0.82	1.05	0.99
18	THieF_4wk-4root	1383.4	2058.0	0.28	0.70	1.08	1.03
19	THieF_3wk-noTransform	1392.9	2008.6	0.38	0.78	1.09	1.01
20	THieF_4wk-noTransform	1397.4	2038.4	0.35	0.77	1.09	1.02
21	THieF_3wk-4root	1401.7	2164.8	0.26	0.80	1.10	1.08
22	sarima_s1-noTransform	1477.9	2162.0	0.37	0.79	1.16	1.08
23	sarima_s7-noTransform	1483.2	2200.6	0.39	0.81	1.16	1.10
24	sarima_s1-4root	1530.2	2227.1	0.26	0.75	1.20	1.12
25	THieF_12wk-noTransform	1553.5	2357.8	0.32	0.76	1.22	1.18
26	sarima_s7-4root	1601.5	2467.2	0.22	0.76	1.25	1.24
27	THieF_12wk-4root	2270.6	2039.9	0.34	0.66	1.78	1.02

Table 3.2. Summary of overall model performance during the validation phase for U.S. national, ordered by ascending WIS

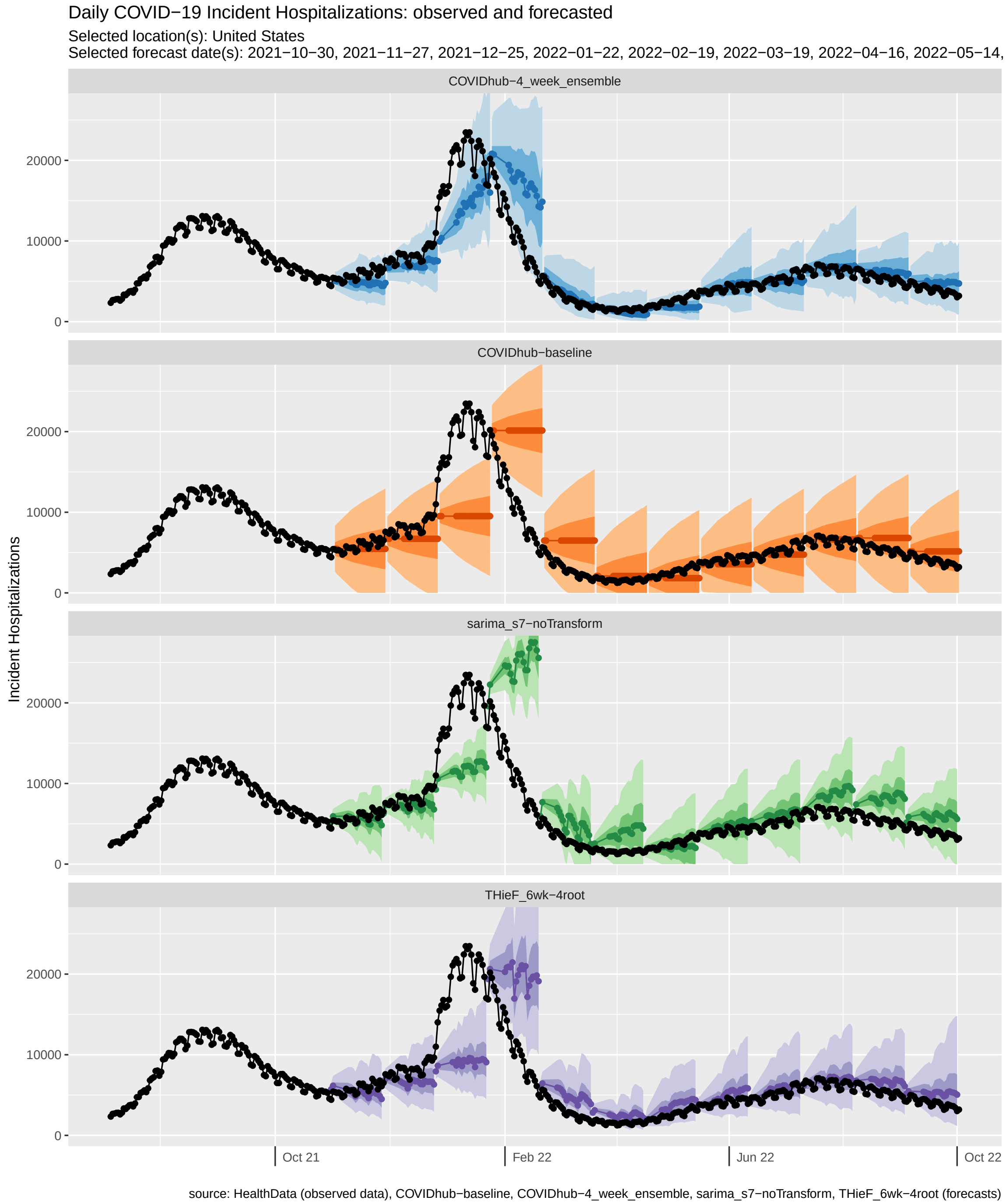

Figure 3.1. U.S. national incident hospitalization forecasts during the testing phase from select models

3.2.1 Overall model performance

Tables 3.3 and 3.4 summarize model performance during the entire testing phase for the averaged states scale and U.S. national scale, respectively. Both geographic scales show very similar model rankings for the overall testing phase evaluation, with the top six models outperforming the COVIDhub-baseline. The COVIDhub-4_week_ensemble has the lowest WIS and MAE by a substantial margin. THieF_6wk-4root display the second lowest WIS and MAE, followed closely behind by THieF_12wk-4root, THieF_ensemble-train3, THieF_ensemble-mean, which all have relative WIS and relative MAE values within 0.02 of each other. The two worst-performing models—sarima_s1-noTransform and sarima_s7-noTransform—are consistent between geographic scales. However, there are some notable differences between performance for averaged states versus U.S. national.

The averaged states results in Table 3.3 display smaller and more closely-clustered WIS and MAE values. Every best subset model is well-calibrated, with near-nominal coverage rates at both uncertainty levels. In fact, our models had better 50% coverage rates than those of the Forecast Hub models. Additionally, the models in the lower half of the ranking, including the COVIDhub-baseline, all showed very similar relative WIS and relative MAE values for this geographic scale.

The U.S. national results in Table 3.4, however, show WIS and MAE values that tend to be more spread out, partially a result of their larger magnitude. At this geographic scale, our models had below-nominal coverage rates for both uncertainty levels. The Forecast Hub models had above-nominal coverage rates at the 50% level but also fell short of nominal coverage at the 95% level. While the THieF ensembles ranked higher than the THieF_12wk-4root model for U.S. national, there was not a substantial difference in terms of relative WIS and relative MAE value. Further, THieF_6wk-noTransform showed more similar performance to the top four ranked models compared to that of the COVIDhub-baseline, unlike for averaged states.

	Model	WIS	MAE	Cov50	Cov95	rWIS	rMAE
1	COVIDhub-4_week_ensemble	30.18	44.87	0.64	0.94	0.72	0.79
2	THieF_6wk-4root	34.55	50.12	0.51	0.90	0.83	0.88
3	THieF_12wk-4root	35.25	51.35	0.50	0.89	0.85	0.90
4	THieF_ensemble-train3	35.62	51.57	0.56	0.91	0.86	0.91
5	THieF_ensemble-mean	35.91	51.80	0.57	0.91	0.86	0.91
6	THieF_6wk-noTransform	39.51	56.87	0.55	0.88	0.95	1.00
7	COVIDhub-baseline	41.63	56.77	0.79	0.96	1.00	1.00
8	sarima_s1-noTransform	42.92	58.04	0.60	0.86	1.03	1.02
9	sarima_s7-noTransform	43.54	58.94	0.60	0.87	1.05	1.04

Table 3.3. Summary of overall model performance during the testing phase for averaged states, ordered by ascending WIS

The sarima_s7-noTransform model, which had the highest WIS and MAE, was substantially worse than the COVIDhub-baseline and the model ranked just above it, sarima_s1-noTransform.

	Model	WIS	MAE	Cov50	Cov95	rWIS	rMAE
1	COVIDhub-4_week_ensemble	1401.3	2085.1	0.63	0.91	0.75	0.83
2	THieF_6wk-4root	1642.5	2358.8	0.40	0.82	0.88	0.94
3	THieF_ensemble-train3	1687.4	2443.4	0.36	0.83	0.90	0.97
4	THieF_ensemble-mean	1700.5	2458.9	0.39	0.84	0.91	0.98
5	THieF_12wk-4root	1723.9	2481.7	0.31	0.81	0.92	0.99
6	THieF_6wk-noTransform	1727.7	2504.8	0.46	0.86	0.92	1.00
7	COVIDhub-baseline	1867.5	2514.1	0.71	0.85	1.00	1.00
8	sarima_s1-noTransform	1949.6	2487.0	0.53	0.82	1.04	0.99
9	sarima_s7-noTransform	2316.8	3042.4	0.41	0.80	1.24	1.21

Table 3.4. Summary of overall model performance during the testing phase for U.S. national, ordered by ascending WIS

3.2.2 Model performance by horizon week

We display the results graphically from this stratified evaluation forward for improved readability, with the full tables of results displayed in the Appendix. Only WIS plots are shown because it is our principle evaluation metric and the MAE plots show very similar results. The by-horizon evaluation (see Figure 3.2 and Supplemental Tables 6 and 8) shows similar model rankings to those of the overall evaluation. The COVIDhub-4_week_ensemble has much lower WIS for all horizons and geographic scales compared to the other eight models while THieF_6wk-4root tends to have the second-lowest WIS for most horizons, beating the corresponding THieF_6wk-noTransform model at every horizon. Additionally, for all but the 1-week ahead horizon, the top six models beat the COVIDhub-baseline in terms of lowest WIS for both geographic scales. The COVIDhub-4_week_ensemble also substantially outperforms all other seven models. However, aggregating the forecasts by horizon illustrates some important differences in accuracy between horizon weeks.

Figure 3.2 shows that for both geographic scales the THieF ensembles outperform the fourth root, original THieF models at shorter horizons while the reverse is true at longer ones. We observe that THieF_12wk-4root has one of the highest WIS values at the 1-week horizon while it has the second lowest WIS at the 4-week horizon. The sarima_s1-noTransform model also shows different relative performance depending on the horizon length, with a slightly higher ranking at the 1-week ahead horizon. At the U.S. national scale, sarima_s7-noTransform shows increasingly worse relative performance compared to the other models as horizon week increases. In general, there are several models that switch ranking order as horizon week increases, shown by crossing lines in Figure 3.2. Further, the COVIDhub-baseline is consistently the third worst model, except for the 1-week ahead horizon.

Coverage rates for the averaged states geographic scale (see Supplemental Table 6) remain high across horizons for all models, with 95% coverage rates fall into no more than 10% of the nominal rate at any horizon. Meanwhile, coverage rates are lower for the U.S. national scale (shown in Supplemental Table 8), falling to up to 19% below the nominal rate (though most are no lower than 15%).

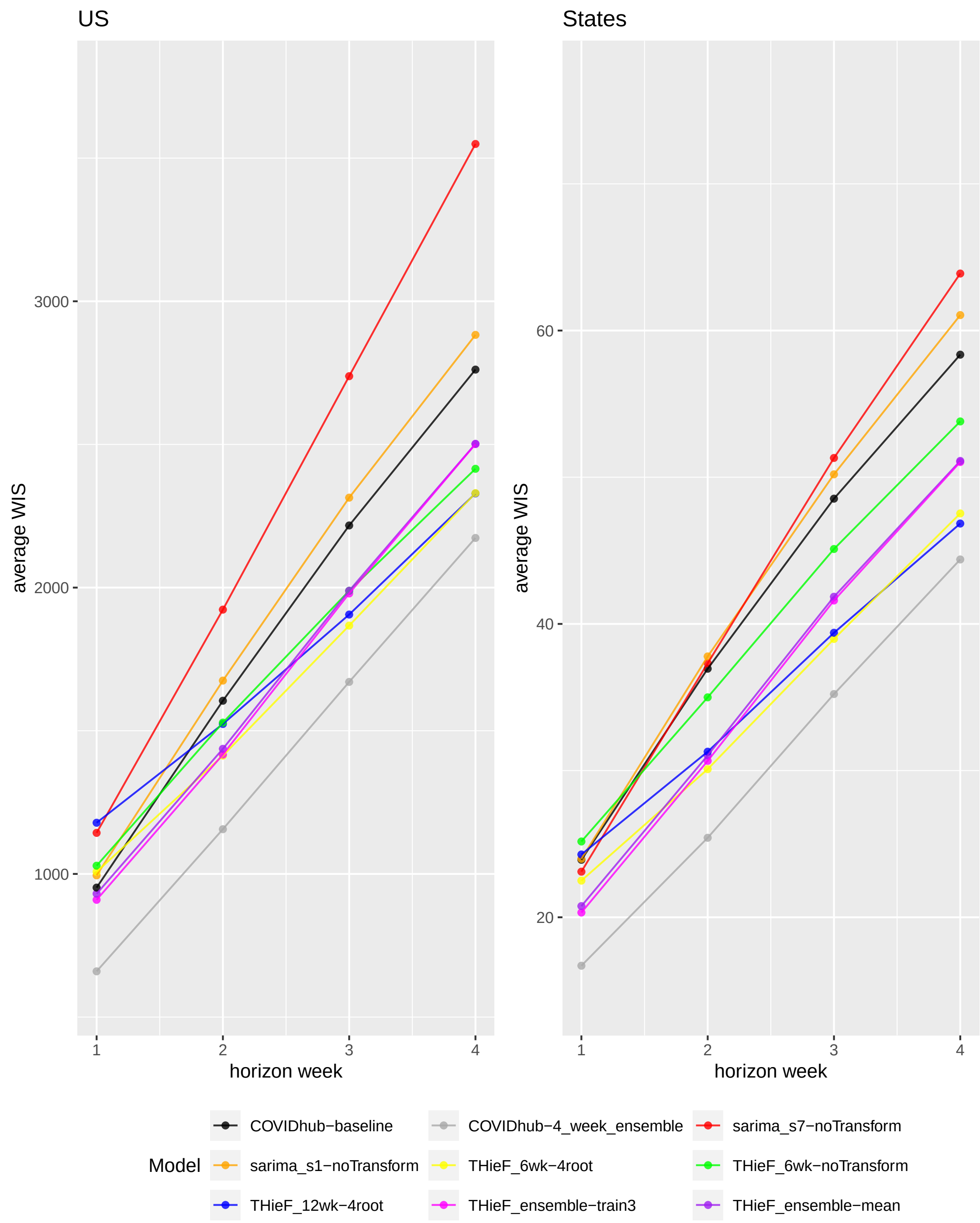

Figure 3.2. Average WIS plotted by horizon week during the testing phase for each model at both geographic scales

3.2.3 Model performance by pandemic wave

One overall theme that continues for the by–pandemic wave evaluation is that the COVIDhub-4_week_ensemble ranks first for lowest WIS and MAE, and this model is substantially better than all the other models at every horizon (see Figure 3.3 and Supplemental Tables 10 and **??**). The next three models in terms of lowest WIS vary for each combination of pandemic wave, geographic scale, and horizon, though the fourth root original THieF models and THieF ensembles usually rank in the top half of models. On the other hand, sarima_s7-noTransform and the COVIDhub-baseline usually rank near the bottom. In Figure 3.3 plots for the same pandemic wave generally bear greater resemblance in terms of overall trends than plots for the same geographic scale; the one notable exception is numerical scale for average WIS.

For the Omicron variant wave at both geographic scales, we see some interesting trends for several models. THieF_12wk-4root has the highest WIS of the nine models at the 1-week ahead horizon but shifts to have one of the lowest WIS values at the 4-week horizon. THieF_6wk-noTransform also shows better relative performance during this wave compared to BA.4/BA.5, especially for the U.S. national scale. Conversely, sarima_s1-noTransform ranks lower among the models for the Omicron wave at all horizon weeks. Once again, we observe a pattern of the THieF ensembles performing better at shorter horizons while the fourth root THieF models perform better at longer ones. All models have greater WIS values during the Omicron wave compared to during the BA.4/BA.5 wave for their respective geographic scale.

For the BA.4/BA.5 variant wave, model rankings by WIS are more distinct and tend to remain static from one horizon week to another, especially for the averaged states scale. This may be a result of this wave being more stable and smaller in magnitude. The best two models—COVIDhub-4_week_ensemble and THieF_6wk-4root—are the same regardless of geographic scale. However, sarima_s1-noTransform and the COVIDhub-baseline rank much higher for the U.S. national scale, with worse performance from THieF_12wk-4root and the two THieF ensembles.

Supplemental Tables 10 and **??** show that for the BA.4/BA.5 wave, coverage rates are exceedingly high for the averaged states level and close to nominal for all but THieF_12wk-4 root for the 4-week ahead horizon at the U.S. national geographic scale. On the other hand, the Omicron wave coverage rates are substantially lower, all nine models failing to reach the nominal level for the 95% prediction interval. Such performance is not unusual given the high magnitude and rapid change seen during this wave.

3.2.4 Model performance by forecast week

For the by–forecast week evaluation, we make four plots for each geographic scale (see Figure 3.4 for averaged states and Figure 3.5 for U.S. national), with WIS or 95% PI coverage (restricted to either 1-week or 4-week ahead) on the y-axis and forecast date on the x-axis. The WIS plots generally follow the shape of the incident hospitalization truth data (see Figure 2.1), with greater WIS observed during times of high incidence and rapid change. The 95% coverage plots tend to follow a vertically flipped transformation of the truth data, with below-nominal coverage rates observed during times of high incidence and rapid change.

Many patterns observed in evaluations described above persisted for this subset of the testing phase analysis. The COVIDhub-4_week_ensemble consistently is one of the the top few, if not the top, models for both WIS and 95% coverage for both geographic scales. The one exception is near the crest of the Omicron wave when this model ranks last or near last for lowest WIS at the 4-week ahead horizon. Here, and in a few other places, the COVIDhub-4_week_ensemble loses out to THieF_6wk-4root and THieF_12wk-4root at the 4-week ahead horizon. The two THieF ensembles also rarely beat the COVIDhub-4_week_ensemble in terms of WIS yet demonstrate

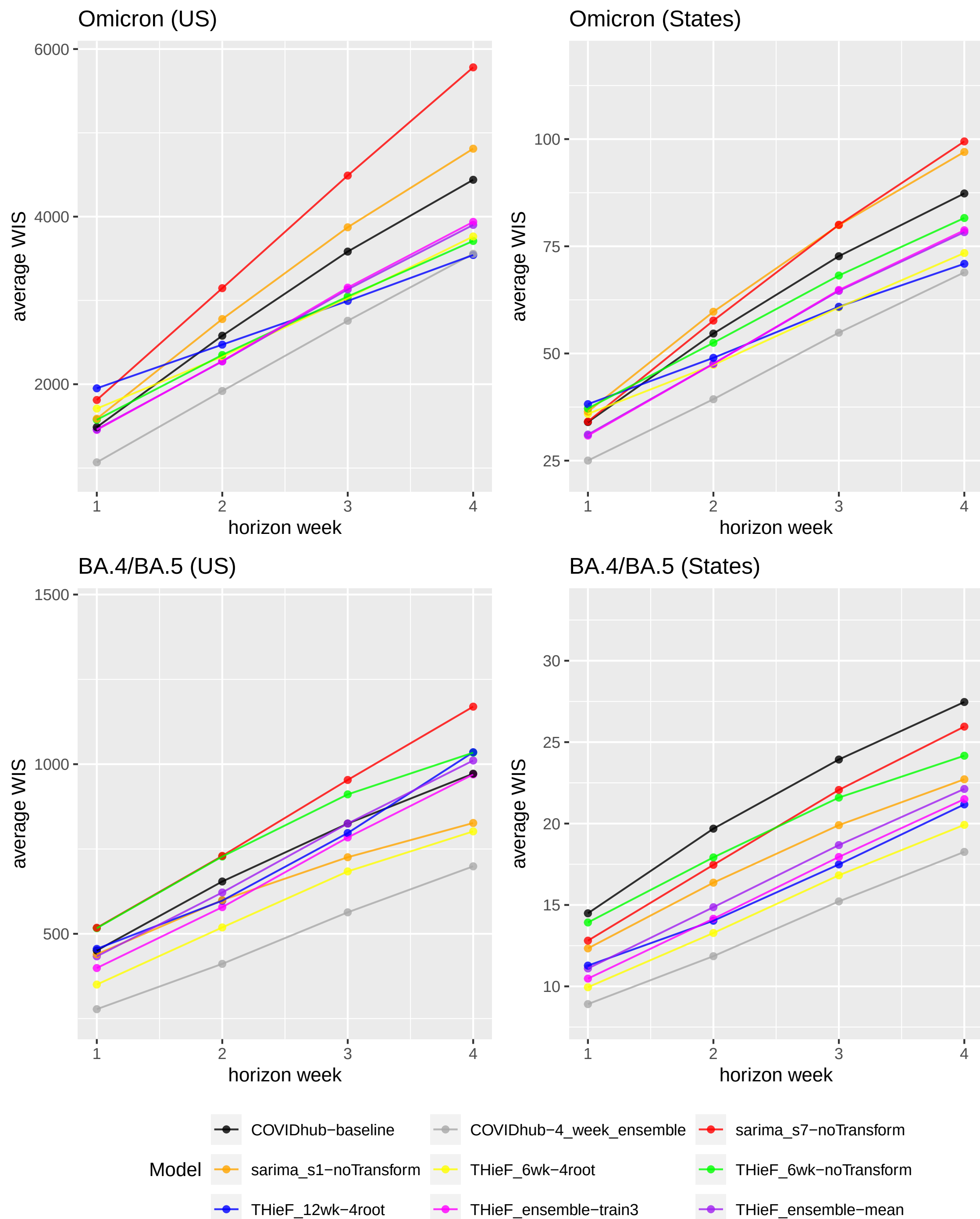

Figure 3.3. Average WIS plotted by pandemic wave and horizon week during the testing phase for each model at both geographic scales

incredibly stable performance, never ranking below sixth. In this way, the THieF ensembles are quite similar to the COVIDhub-baseline during times of high incidence and rapid change: their WIS is more stable and consistent from week-to-week than that of other models; however, THieF_ensemble-mean and THieF_ensemble-train3 often out-perform the COVIDhub-baseline, which has the worst WIS for several weeks. Generally, though, sarima_s1-noTransform and sarima_s7-noTransform alternate for having the worst WIS and 95% coverage rates. The one notable exception to this is leading up to the Omicron peak, in which the three original THieF models had the highest WIS for a single week at both geographic scales.

The by–forecast week break down of THieF_6wk-noTransform is particularly interesting. This model tended to fall in the middle of the rankings for the previously described evaluations, yet Figures 3.4 and 3.5 show that this model usually exhibited more extreme performance, either performing rather well or quite badly. Investigating less aggregate scores for THieF_12wk-4root similarly reveals several instances of good performance, usually at long horizons under settings of rapid change and/or high incidence. In addition, we notice that sarima_s7-noTransform demonstrated noticeably worse performance for U.S. national WIS at a horizon of 1 week compared to that for averaged states. Further, both fourth root original THieF display strangely low 95% coverage for averaged states following the decline of the Omicron wave at both plotted horizons, especially since the other seven models experience significantly higher coverage rates during this time. (Comparisons among models for U.S. national 95% coverage by forecast week are harder to make since this location scale only includes seven 95% interval forecasts for one location (compared to seven per 52 locations for averaged states), so we simply allow the plots in Figure 3.4 to speak for themselves.)

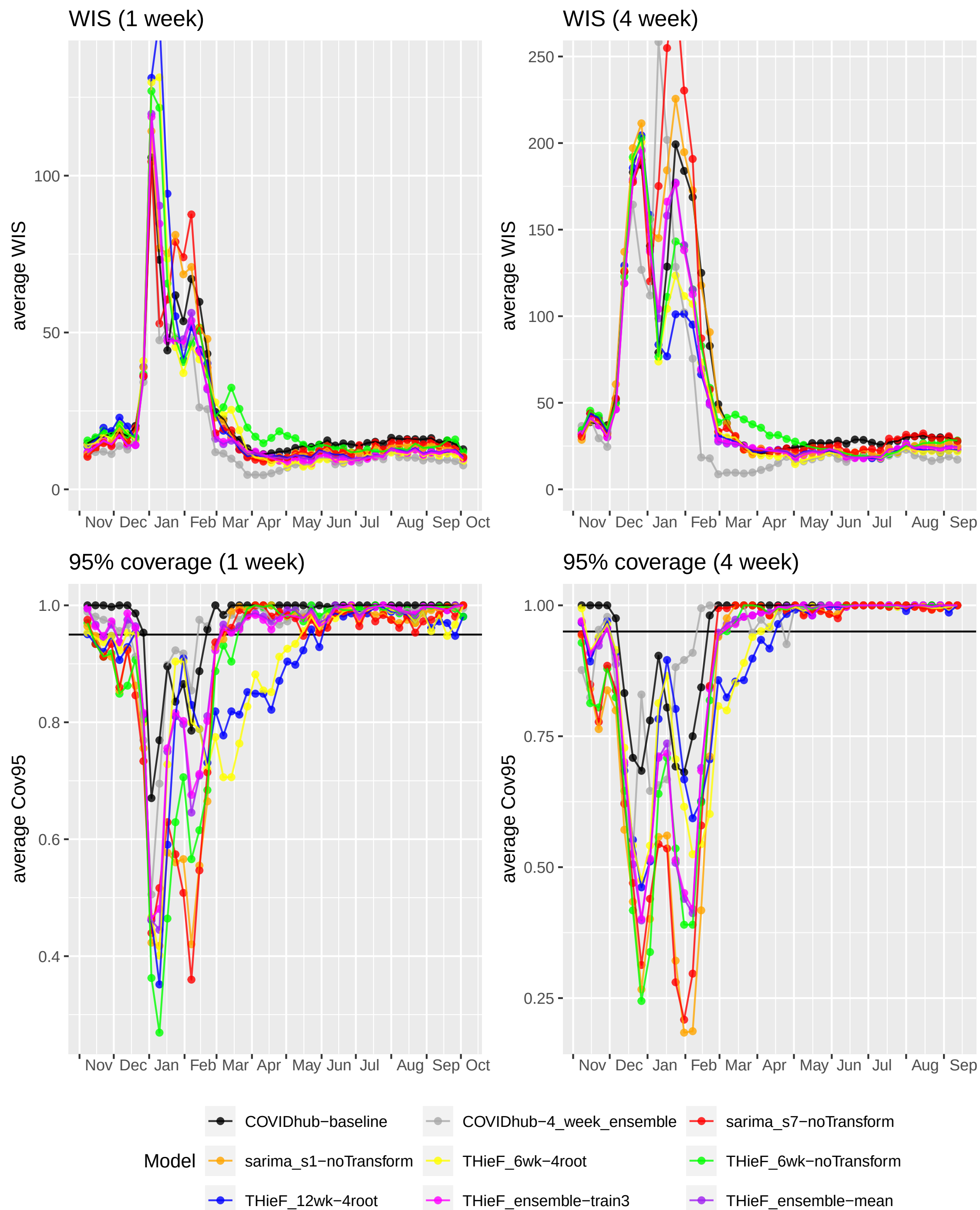

Figure 3.4. Average h-week ahead WIS and 95% PI coverage plotted by forecast week during the testing phase for each model for averaged states

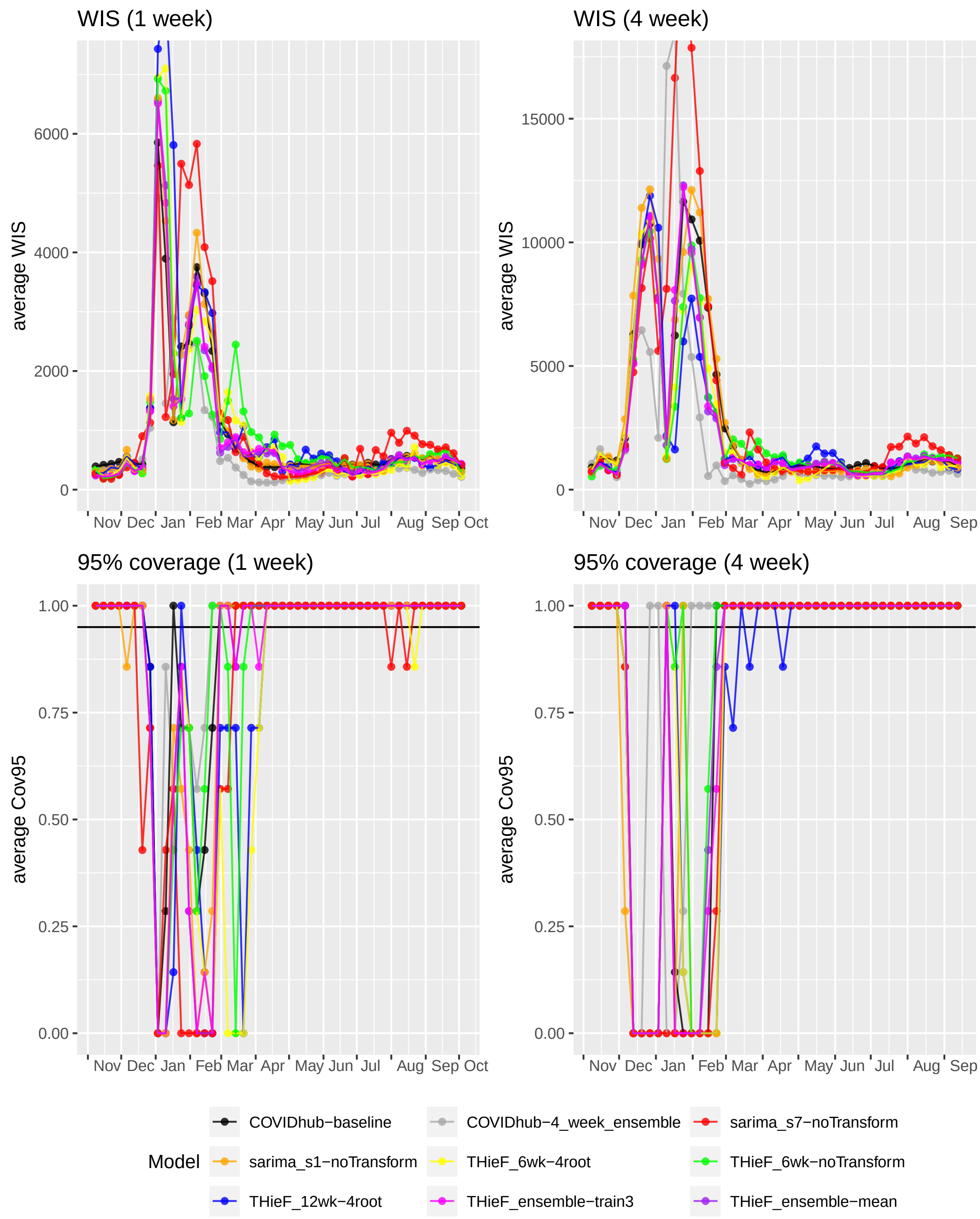

Figure 3.5. Average h-week ahead WIS and 95% PI coverage plotted by forecast week during the testing phase for each model for U.S. national

3.2.5 Model performance by location

Figure 3.6 summarizes the by-location evaluation, with the models on the x-axis ordered by lowest-to-highest overall relative WIS at the averaged states location scale for the testing phase and locations on the y-axis ordered by descending cumulative hospitalizations. The top six models show either better or equal performance than the COVIDhub-baseline—represented by shades of blue ranging from dark to light and white, respectively—with the COVIDhub-4_week_ensemble performing the best. The models ranked second, fourth, and fifth demonstrate fairly close WIS values across all of the locations, though THieF_6wk-4root is marginally better with lower relative WIS by a difference of more than 0.05 in over 18% of locations. On the other hand, THieF_12wk-4root, which ranked third in terms of lowest WIS, displays a slightly different set of locations that it forecasted well compared to the other top models, as can be seen by the different colors and shading for its column in Figure 3.6.

The worst two models include some shades of red which represent performance worse than the COVIDhub-baseline. However, it should be noted that these models actually perform better than the baseline for many locations. In addition, most of the worst performance relative to the baseline occurs in the locations with the highest cumulative hospitalizations during the testing phase, except for the THieF_12wk-4root model with low count locations New Hampshire and Vermont. The general trend, though, is perhaps unsurprising since we know that high-count settings tend to be more difficult to forecast accurately.

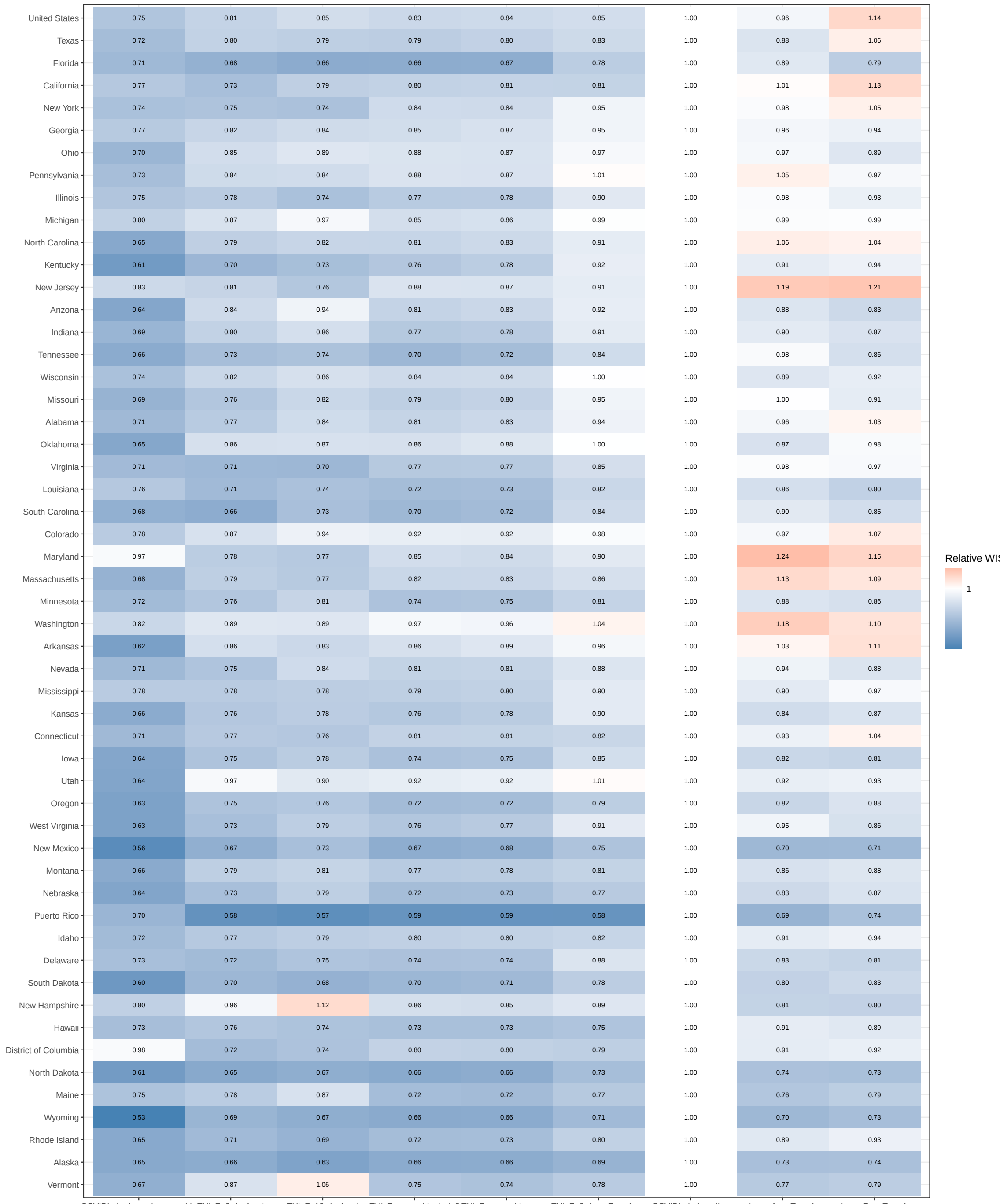

Figure 3.6. Relative WIS plotted by location during the testing phase for each model across all horizons

CHAPTER 4

DISCUSSION

THieF is a unique forecast methodology that integrates hierarchical forecasting with forecast combination and shows a number of improvements in forecast accuracy over the COVIDhub-baseline. Such accuracy gains vary based on each model's temporal hierarchy and data transformation (or lack of thereof), with well-crafted THieF models demonstrating competitive performance. The best model, THieF_6wk-4root, almost always ranked second in terms of WIS and MAE with typically nearer-nominal coverage rates for both uncertainty levels. This model produced consistently accurate forecasts for incident hospitalizations using real-time truth data during the 48-week long testing phase. Other well-performing models, like THieF_12wk-4root, THieF_ensemble-train3, and THieF_ensemble-mean, also produced competitive forecasts with substantial gains over the COVIDhub-baseline model. The testing phase evaluations allow us to draw the following conclusions about the models we generated and the underlying methodology.

- **The THieF ensembles and original THieF models tend to be accurate compared to baseline model.** In terms of WIS and MAE, THieF_6wk-4root,THieF_12wk-4root, THieF_ensemble-train3, and THieF_ensemble-mean almost always beat the COVIDhub-baseline, while THieF_6wk-noTransform almost always performed as well (though sometimes better) than the baseline. This was true across evaluation split by horizon, pandemic wave, forecast date, and location; for the top three models, approximately equal improvements occurred in stable, low incidence settings and rapidly-changing, high-incidence settings. These models also demonstrated good 50% coverage and reasonable 95% coverage, with narrower (and thus typically more useful) prediction intervals. Hence, we conclude that these four models may be presented as better alternatives to the COVIDHub-baseline for incident hospitalizations.

- **THieF_6wk-4root, followed closely by THieF_12wk-4root and the THieF ensembles, yields the best performance when compared to alternate models from the best subset.** THieF_6wk-4root, THieF_12wk-4root, THieF_ensemble-train3, and THieF_ensemble-mean ranked within the top five for most testing phase evaluation, which compared nine models including the COVIDhub-baseline and COVIDhub-4_week_ensemble. In fact, except for the 1-week ahead horizon during times of rapid change or high incidence, THieF_6wk-4root ranked second almost every analysis. It should be noted, however, that despite slightly lower rankings, especially during times of high incidence or rapid change, the THieF ensembles showed greater stability and resistance to jumps in WIS, MAE, and PI coverage compared to THieF_6wk-4root. Overall, these models consistently had competitive performance for all four metrics out of the best subset models that moved on from the validation phase; therefore, depending on the decision-making involved needs involved, we conclude that these models would be suitable for informing to COVID-19.

- **The accuracy improvements of the original THieF models over the COVIDhub-baseline are likely a result of the THieF methodology itself, not the underlying ARIMA base forecaster.** During the analyses of both phases, sarima_s1-noTransform and sarima_s7-noTransform often demonstrated poor performance, as did the other ARIMA and SARIMA models in the validation phase, and only rarely beat the COVIDhub-baseline. The best performance sarima_s1-noTransform achieved occurred during static, low-count periods at short horizons, which was equivalent to that of higher ranking models. We thus conclude that it is not the underlying ARIMA base forecaster that lends most THieF models their accuracy improvements but rather the other aspects of the THieF methodology. Namely, the union of hierarchical forecasting and forecast combination.

- **There may not be a meaningful difference between the trained and untrained THieF ensembles.** THieF_ensemble-train3 and THieF_ensemble-mean showed limited differences in performance during the testing phase analysis, regardless of location scale, horizon, pandemic wave, forecast date, or evaluation metric. THieF_ensemble-train3 is marginally better than the untrained THieF_ensemble-mean in most evaluations, the largest difference in relative WIS being 0.07 for any single horizon, horizon–wave, or location comparison between the models. Usually, though, the relative WIS of these two models is much closer. The by–forecast date plots provide further evidence of their similarity, as the two ensembles have almost identical WIS for every forecast date (see Figures 3.4 and 3.5). Since this trained ensemble has a small θ value, indicating a lighter weighting scheme, this result is not all too surprising.

- **The fourth root data transformation is essential to achieving the best results when using the THieF methodology.** We saw a range of performance demonstrated by the different THieF models during the validation and testing phase analyses, with THieF_6wk-4root and THieF_12wk-4root consistently obtaining some of the best WIS values. Conversely, THieF_6wk-noTransform had more middling performance and almost always was worse than THieF_6wk-4root. Although we did not pass on THieF_12wk-noTransform, this model ranked below its fourth root counterpart for the last 17 weeks[1] of the validation phase. Other THieF models also out-performed the COVIDhub-baseline during the validation phase, including the THieF models with the second longest top aggregation level of eight weeks. Hence, it appears that satisfying the constant variance assumption of the underlying ARIMA model substantially improves accuracy gains from the THieF methodology.

Our results show that THieF is a powerful methodology that can model the real-world complexities of an infectious disease system, particularly if data transformations are used to obtain approximately constant variance. The initial base forecasts are created using a classic statistical model, a type of model that has shown good performance when forecasting for short horizons in outbreak settings[13], but it is the final reconciliation of these forecasts that likely explains THieF's accuracy gains. Ensembling a large number of THieF models with a range of top aggregation levels significantly added to initial accuracy improvements, but ultimately is not better than THieF models with a fourth root transformation. Finally, careful selection of the period of analysis and day of the week on which to make forecasts helped to prevent any potential issues due to missing data, while the choice of incident hospitalizations as the forecast target limited in accuracies due to under reporting.

Further work might consider different evaluation metrics, alternative ways of aggregating forecast scores, implementing logarithmic scores, or using metrics that are not scale-dependent. The magnitude of WIS and MAE is dependent on the scale of forecasts, meaning that averages calculated from these metrics will be dominated by forecasts with large WIS and MAE values. For example, average metrics calculated during the entire validation phase will have the largest contributions from longer hori-

[1]We focus on the last 17 week evaluation rather than the overall one for the validation phase because the latter analysis included an anomaly when THieF_12wk-4root had an unusually large upper limit for many of its interval forecasts, which hid its otherwise better performance.

zons and high-count locations; that is, a model that forecasts low-count locations well but high-count locations poorly is penalized in the overall evaluation more harshly than that of a model that does the opposite. The same is true for short versus long horizons. We attempted to address this issue by considering a range of stratified analyses, but these evaluations are limited in scope. Future work that scores forecasts using other metrics that are not scale-dependent may better solve the problem of unequal contributions to aggregated scores.

The other main direction for future work is obtaining properly calibrated reconciled forecasts either through sampling or by employing one of the existing probabilistic applications of THieF to forecast COVID-19 incident hospitalizations. While we generally did not observe great deviations from nominal-level coverage in our work, the THieF models coverage rates could be improved and increase overall model accuracy. Using a probabilistic implementation of THieF would generate a complete probabilistic distribution instead of using prediction intervals from point forecasts, which may be more useful or more accurate, especially in terms of improving coverage rates.

Alternative ensemble formulations might also be considered, such as a median ensemble, a trained ensemble with a shifting θ value, ensembles of only THieF models with a fourth root data transform, or even ensembles of THieF implementations specifically tailored for point and probabilistic forecasting. We are also curious about accuracy comparisons to other models that submitted to the Forecast Hub during the period of analysis.

We demonstrated the utility and success of THieF and ensembles of THieF models when forecasting incident hospitalizations for COVID-19 through the results of this paper. THieF models made with fourth root–transformed data and THieF ensembles demonstrate clear improvements over the baseline model for both point and probabilistic forecasts, with THieF_6wk-4root ranking just below the top-performing COVIDhub-4_week_ensemble model. Infectious disease forecasting continues to be important for acute disease outbreaks, epidemics, pandemic, and endemic diseases, and we hope that the THieF methodology will be used in such settings given its demonstrated accuracy improvements in this application.

APPENDIX: SUPPLEMENTAL MATERIALS

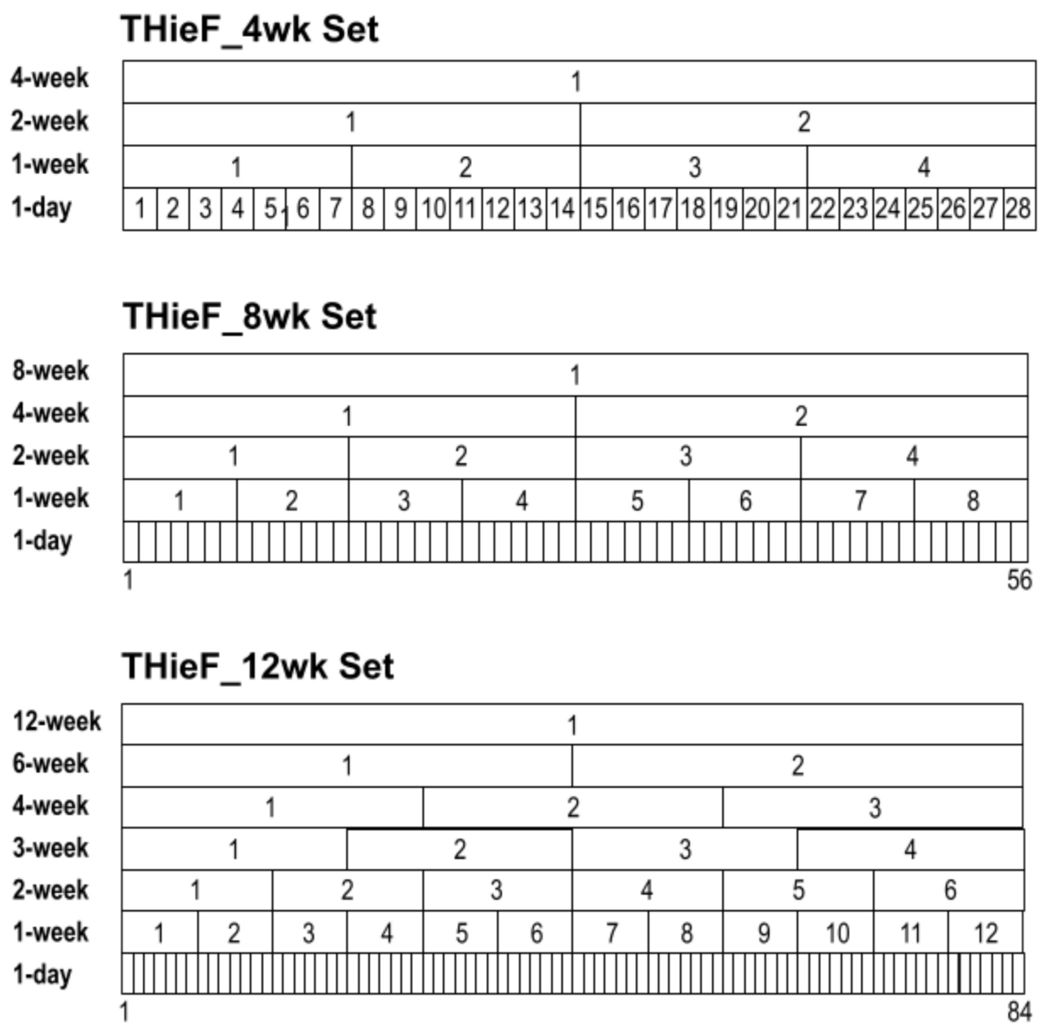

Figure 1. Illustration of three hierarchy schemes used to create six of the fourteen total original THieF models

Table 2. Summary of model performance during the last 17 weeks of the validation phase for averaged states, ordered by ascending WIS

Model	WIS	MAE	Cov50	Cov95	rWIS	rMAE
THieF_ensemble-train6.5	32.656	49.713	0.517	0.921	0.763	0.772
THieF_ensemble-train3	32.676	49.547	0.521	0.920	0.764	0.769
THieF_ensemble-train10	32.699	49.867	0.514	0.921	0.764	0.774
THieF_ensemble-mean	32.708	49.190	0.527	0.921	0.764	0.764
THieF_ensemble-train1	32.734	49.462	0.525	0.920	0.765	0.768
THieF_ensemble-train15	32.827	50.106	0.507	0.920	0.767	0.778
THieF_ensemble-train20	32.990	50.345	0.502	0.917	0.771	0.782
THieF_ensemble-train25	33.148	50.564	0.498	0.914	0.775	0.785
THieF_6wk-4root	33.361	49.308	0.491	0.909	0.780	0.766
THieF_2wk-4root	34.171	51.407	0.487	0.902	0.799	0.798
THieF_2wk-noTransform	34.358	50.181	0.514	0.879	0.803	0.779
THieF_6wk-noTransform	34.721	51.211	0.512	0.884	0.812	0.795
THieF_12wk-4root	34.898	50.270	0.488	0.905	0.816	0.780
THieF_8wk-4root	35.054	51.488	0.480	0.897	0.819	0.799
THieF_1wk-4root	35.120	53.469	0.474	0.896	0.821	0.830
THieF_8wk-noTransform	35.334	52.163	0.502	0.878	0.826	0.810
THieF_3wk-4root	36.397	54.575	0.480	0.888	0.851	0.847
THieF_3wk-noTransform	36.981	53.938	0.493	0.869	0.864	0.837
THieF_4wk-4root	36.988	54.772	0.473	0.884	0.865	0.850
THieF_1wk-noTransform	37.223	54.871	0.481	0.866	0.870	0.852
THieF_4wk-noTransform	38.179	55.999	0.480	0.861	0.892	0.869
THieF_12wk-noTransform	38.287	53.261	0.501	0.856	0.895	0.827
sarima_s1-4root	41.743	62.340	0.432	0.872	0.976	0.968
COVIDhub-baseline	42.785	64.408	0.716	0.969	1.000	1.000
sarima_s7-4root	43.511	64.229	0.439	0.872	1.017	0.997
sarima_s7-noTransform	43.516	62.037	0.459	0.840	1.017	0.963
sarima_s1-noTransform	46.119	66.363	0.435	0.836	1.078	1.030

Table 4. Summary of model performance during the last 17 weeks of the validation phase for U.S. national, ordered by ascending WIS

Model	WIS	MAE	Cov50	Cov95	rWIS	rMAE
THieF_12wk-4root	1134.2	1632.2	0.58	0.84	0.73	0.63
THieF_ensemble-mean	1168.8	1887.3	0.36	0.90	0.75	0.72
THieF_ensemble-train1	1191.0	1932.5	0.36	0.90	0.76	0.74
THieF_6wk-4root	1198.9	1799.3	0.47	0.86	0.77	0.69
THieF_ensemble-train3	1210.1	1970.1	0.34	0.91	0.78	0.76
THieF_6wk-noTransform	1216.0	1906.2	0.34	0.86	0.78	0.73
THieF_ensemble-train6.5	1249.3	2033.2	0.33	0.92	0.80	0.78
THieF_8wk-noTransform	1283.1	2045.7	0.36	0.86	0.82	0.78
THieF_ensemble-train10	1293.2	2097.3	0.32	0.92	0.83	0.80
THieF_ensemble-train15	1356.6	2195.9	0.31	0.91	0.87	0.84
THieF_12wk-noTransform	1376.6	1953.7	0.54	0.78	0.88	0.75
THieF_1wk-noTransform	1409.5	2240.3	0.27	0.80	0.91	0.86
THieF_ensemble-train20	1411.9	2283.5	0.31	0.90	0.91	0.88
THieF_8wk-4root	1425.9	2111.2	0.33	0.78	0.92	0.81
THieF_4wk-noTransform	1450.7	2141.7	0.34	0.67	0.93	0.82
THieF_ensemble-train25	1455.3	2349.6	0.30	0.89	0.94	0.90
THieF_2wk-4root	1456.7	2291.5	0.26	0.83	0.94	0.88
THieF_2wk-noTransform	1457.8	2180.4	0.31	0.73	0.94	0.84
THieF_3wk-noTransform	1487.5	2163.4	0.32	0.66	0.96	0.83
THieF_3wk-4root	1533.0	2428.1	0.34	0.84	0.98	0.93
COVIDhub-baseline	1555.9	2606.5	0.37	0.97	1.00	1.00
THieF_1wk-4root	1565.4	2542.2	0.27	0.85	1.01	0.97
THieF_4wk-4root	1614.4	2433.5	0.30	0.71	1.04	0.93
sarima_s1-noTransform	1774.3	2639.2	0.22	0.71	1.14	1.01
sarima_s1-4root	1803.2	2614.4	0.22	0.75	1.16	1.00
sarima_s7-noTransform	1922.0	2884.6	0.15	0.65	1.24	1.11
sarima_s7-4root	2155.2	3344.1	0.17	0.73	1.38	1.28

Model	hzn	WIS	MAE	Cov50	Cov95	rWIS	rMAE
COVIDhub-4_week_ensemble	1	16.70	25.11	0.65	0.96	0.70	0.79
THieF_ensemble-train3	1	20.32	30.04	0.55	0.93	0.85	0.94
THieF_ensemble-mean	1	20.76	30.52	0.56	0.93	0.87	0.96
THieF_6wk-4root	1	22.50	32.97	0.49	0.89	0.94	1.03
sarima_s7-noTransform	1	23.11	31.92	0.57	0.89	0.97	1.00
COVIDhub-baseline	1	23.92	31.87	0.80	0.97	1.00	1.00
sarima_s1-noTransform	1	23.98	32.81	0.60	0.89	1.00	1.03
THieF_12wk-4root	1	24.28	35.21	0.48	0.89	1.01	1.10
THieF_6wk-noTransform	1	25.17	36.41	0.52	0.89	1.05	1.14
COVIDhub-4_week_ensemble	2	25.42	37.90	0.66	0.95	0.69	0.77
THieF_6wk-4root	2	30.10	43.94	0.52	0.91	0.81	0.89
THieF_ensemble-train3	2	30.66	44.49	0.56	0.91	0.83	0.90
THieF_ensemble-mean	2	31.04	44.80	0.58	0.91	0.84	0.90
THieF_12wk-4root	2	31.29	45.58	0.51	0.90	0.85	0.92
THieF_6wk-noTransform	2	34.99	50.27	0.56	0.88	0.95	1.02
COVIDhub-baseline	2	36.95	49.48	0.80	0.96	1.00	1.00
sarima_s7-noTransform	2	37.32	50.28	0.60	0.87	1.01	1.02
sarima_s1-noTransform	2	37.78	50.84	0.61	0.87	1.02	1.03
COVIDhub-4_week_ensemble	3	35.22	52.52	0.64	0.94	0.73	0.79
THieF_6wk-4root	3	38.98	56.36	0.52	0.90	0.80	0.85
THieF_12wk-4root	3	39.40	57.34	0.51	0.90	0.81	0.86
THieF_ensemble-train3	3	41.60	59.88	0.56	0.90	0.86	0.90
THieF_ensemble-mean	3	41.85	60.07	0.58	0.90	0.86	0.91
THieF_6wk-noTransform	3	45.11	64.83	0.56	0.87	0.93	0.98
COVIDhub-baseline	3	48.54	66.30	0.79	0.95	1.00	1.00
sarima_s1-noTransform	3	50.20	67.64	0.60	0.85	1.03	1.02
sarima_s7-noTransform	3	51.31	69.07	0.60	0.86	1.06	1.04
COVIDhub-4_week_ensemble	4	44.39	65.43	0.63	0.93	0.76	0.80
THieF_12wk-4root	4	46.84	68.49	0.50	0.89	0.80	0.84
THieF_6wk-4root	4	47.53	68.48	0.52	0.88	0.81	0.84
THieF_ensemble-train3	4	51.04	73.47	0.55	0.89	0.88	0.90
THieF_ensemble-mean	4	51.11	73.39	0.56	0.88	0.88	0.90
THieF_6wk-noTransform	4	53.80	77.46	0.55	0.86	0.92	0.95
COVIDhub-baseline	4	58.35	81.23	0.77	0.95	1.00	1.00

Table 6. Summary of model performance during the testing phase stratified by horizon week for averaged states, ordered by ascending WIS

sarima_s1-noTransform	4	61.05	82.67	0.58	0.83	1.05	1.02
sarima_s7-noTransform	4	63.89	86.44	0.60	0.85	1.09	1.06

Model	hzn	WIS	MAE	Cov50	Cov95	rWIS	rMAE
COVIDhub-4_week_ensemble	1	659.8	966.8	0.68	0.94	0.69	0.77
THieF_ensemble-train3	1	909.6	1348.8	0.37	0.86	0.95	1.07
THieF_ensemble-mean	1	930.9	1372.9	0.38	0.86	0.98	1.09
COVIDhub-baseline	1	952.2	1258.8	0.73	0.92	1.00	1.00
sarima_s1-noTransform	1	995.0	1308.1	0.59	0.87	1.04	1.04
THieF_6wk-4root	1	1007.4	1419.1	0.45	0.79	1.06	1.13
THieF_6wk-noTransform	1	1028.7	1467.4	0.46	0.88	1.08	1.17
sarima_s7-noTransform	1	1143.4	1533.9	0.32	0.81	1.20	1.22
THieF_12wk-4root	1	1178.7	1666.8	0.33	0.83	1.24	1.32
COVIDhub-4_week_ensemble	2	1156.2	1694.7	0.66	0.92	0.72	0.80
THieF_6wk-4root	2	1413.2	2029.0	0.42	0.85	0.88	0.96
THieF_ensemble-train3	2	1417.1	2046.9	0.39	0.84	0.88	0.97
THieF_ensemble-mean	2	1436.9	2070.8	0.40	0.85	0.90	0.98
THieF_12wk-4root	2	1523.7	2185.5	0.37	0.82	0.95	1.03
THieF_6wk-noTransform	2	1528.5	2177.0	0.47	0.86	0.95	1.03
COVIDhub-baseline	2	1604.9	2113.3	0.74	0.85	1.00	1.00
sarima_s1-noTransform	2	1675.2	2133.3	0.56	0.83	1.04	1.01
sarima_s7-noTransform	2	1923.1	2536.3	0.41	0.82	1.20	1.20
COVIDhub-4_week_ensemble	3	1670.6	2528.8	0.60	0.90	0.75	0.84
THieF_6wk-4root	3	1867.4	2715.3	0.37	0.84	0.84	0.91
THieF_12wk-4root	3	1905.9	2766.1	0.31	0.80	0.86	0.92
THieF_ensemble-train3	3	1979.6	2845.1	0.35	0.81	0.89	0.95
THieF_6wk-noTransform	3	1989.0	2883.6	0.46	0.85	0.90	0.96
THieF_ensemble-mean	3	1989.0	2860.1	0.39	0.83	0.90	0.95
COVIDhub-baseline	3	2217.0	2993.7	0.71	0.83	1.00	1.00
sarima_s1-noTransform	3	2314.1	2946.7	0.49	0.81	1.04	0.98
sarima_s7-noTransform	3	2738.4	3581.4	0.46	0.80	1.24	1.20
COVIDhub-4_week_ensemble	4	2173.4	3232.7	0.56	0.89	0.79	0.85
THieF_12wk-4root	4	2328.9	3370.2	0.22	0.78	0.84	0.89
THieF_6wk-4root	4	2330.0	3341.7	0.36	0.80	0.84	0.88
THieF_6wk-noTransform	4	2414.9	3567.4	0.43	0.85	0.88	0.94
THieF_ensemble-train3	4	2501.1	3615.3	0.32	0.80	0.91	0.96
THieF_ensemble-mean	4	2502.2	3613.6	0.38	0.80	0.91	0.95
COVIDhub-baseline	4	2761.4	3782.1	0.67	0.80	1.00	1.00

Table 8. Summary of model performance during the testing phase stratified by horizon week for U.S. national, ordered by ascending WIS

sarima_s1-noTransform	4	2882.6	3644.9	0.46	0.78	1.04	0.96
sarima_s7-noTransform	4	3549.4	4630.4	0.45	0.76	1.28	1.22

Model	hzn	wave	WIS	MAE	Cov50	Cov95	rWIS	rMAE
COVIDhub-4_week_ensemble	1	ba4_ba5	8.91	13.43	0.72	0.99	0.61	0.81
THieF_6wk-4root	1	ba4_ba5	9.95	14.83	0.62	0.97	0.69	0.89
THieF_ensemble-train3	1	ba4_ba5	10.48	15.62	0.67	0.99	0.72	0.94
THieF_ensemble-mean	1	ba4_ba5	11.10	16.47	0.69	0.99	0.77	0.99
THieF_12wk-4root	1	ba4_ba5	11.28	16.92	0.60	0.96	0.78	1.02
sarima_s1-noTransform	1	ba4_ba5	12.33	16.95	0.74	0.98	0.85	1.02
sarima_s7-noTransform	1	ba4_ba5	12.81	19.00	0.68	0.98	0.88	1.14
THieF_6wk-noTransform	1	ba4_ba5	13.93	20.54	0.70	0.99	0.96	1.24
COVIDhub-baseline	1	ba4_ba5	14.49	16.59	0.92	1.00	1.00	1.00
COVIDhub-4_week_ensemble	2	ba4_ba5	11.85	17.64	0.76	1.00	0.60	0.81
THieF_6wk-4root	2	ba4_ba5	13.28	19.07	0.69	0.99	0.68	0.88
THieF_12wk-4root	2	ba4_ba5	14.03	20.32	0.68	0.98	0.71	0.94
THieF_ensemble-train3	2	ba4_ba5	14.15	20.34	0.74	0.99	0.72	0.94
THieF_ensemble-mean	2	ba4_ba5	14.86	21.20	0.75	0.99	0.76	0.98
sarima_s1-noTransform	2	ba4_ba5	16.37	22.05	0.79	0.99	0.83	1.02
sarima_s7-noTransform	2	ba4_ba5	17.46	25.07	0.75	0.99	0.89	1.16
THieF_6wk-noTransform	2	ba4_ba5	17.92	25.69	0.77	1.00	0.91	1.19
COVIDhub-baseline	2	ba4_ba5	19.69	21.67	0.97	1.00	1.00	1.00
COVIDhub-4_week_ensemble	3	ba4_ba5	15.22	22.86	0.76	0.99	0.64	0.83
THieF_6wk-4root	3	ba4_ba5	16.82	23.30	0.72	0.99	0.70	0.85
THieF_12wk-4root	3	ba4_ba5	17.49	25.05	0.71	0.99	0.73	0.91
THieF_ensemble-train3	3	ba4_ba5	17.95	25.29	0.76	1.00	0.75	0.92
THieF_ensemble-mean	3	ba4_ba5	18.68	26.22	0.78	1.00	0.78	0.95
sarima_s1-noTransform	3	ba4_ba5	19.90	27.30	0.80	0.99	0.83	0.99
THieF_6wk-noTransform	3	ba4_ba5	21.59	31.20	0.78	1.00	0.90	1.14
sarima_s7-noTransform	3	ba4_ba5	22.06	31.34	0.78	0.99	0.92	1.14
COVIDhub-baseline	3	ba4_ba5	23.93	27.49	0.97	1.00	1.00	1.00
COVIDhub-4_week_ensemble	4	ba4_ba5	18.26	27.51	0.76	0.99	0.66	0.84
THieF_6wk-4root	4	ba4_ba5	19.92	26.57	0.75	1.00	0.72	0.81
THieF_12wk-4root	4	ba4_ba5	21.18	29.92	0.71	0.99	0.77	0.91
THieF_ensemble-train3	4	ba4_ba5	21.50	29.63	0.78	1.00	0.78	0.90
THieF_ensemble-mean	4	ba4_ba5	22.13	30.41	0.79	1.00	0.81	0.93
sarima_s1-noTransform	4	ba4_ba5	22.72	31.82	0.81	0.99	0.83	0.97
THieF_6wk-noTransform	4	ba4_ba5	24.17	34.33	0.80	1.00	0.88	1.05

sarima_s7-noTransform	4	ba4_ba5	25.95	36.72	0.80	0.99	0.94	1.12
COVIDhub-baseline	4	ba4_ba5	27.46	32.80	0.97	1.00	1.00	1.00
COVIDhub-4_week_ensemble	1	omicron	25.02	37.57	0.57	0.92	0.74	0.78
THieF_ensemble-train3	1	omicron	30.84	45.43	0.41	0.86	0.91	0.94
THieF_ensemble-mean	1	omicron	31.09	45.53	0.42	0.86	0.91	0.94
COVIDhub-baseline	1	omicron	34.00	48.18	0.66	0.94	1.00	1.00
sarima_s7-noTransform	1	omicron	34.12	45.73	0.46	0.79	1.00	0.95
THieF_6wk-4root	1	omicron	35.91	52.36	0.35	0.81	1.06	1.09
sarima_s1-noTransform	1	omicron	36.43	49.76	0.46	0.79	1.07	1.03
THieF_6wk-noTransform	1	omicron	37.18	53.36	0.34	0.79	1.09	1.11
THieF_12wk-4root	1	omicron	38.17	54.74	0.35	0.81	1.12	1.14
COVIDhub-4_week_ensemble	2	omicron	39.33	58.67	0.56	0.91	0.72	0.75
THieF_6wk-4root	2	omicron	47.34	69.42	0.34	0.82	0.87	0.89
THieF_ensemble-train3	2	omicron	47.58	69.23	0.39	0.83	0.87	0.89
THieF_ensemble-mean	2	omicron	47.61	68.98	0.40	0.83	0.87	0.88
THieF_12wk-4root	2	omicron	48.98	71.48	0.34	0.82	0.90	0.92
THieF_6wk-noTransform	2	omicron	52.49	75.46	0.35	0.77	0.96	0.97
COVIDhub-baseline	2	omicron	54.64	77.98	0.64	0.92	1.00	1.00
sarima_s7-noTransform	2	omicron	57.67	76.11	0.45	0.76	1.05	0.98
sarima_s1-noTransform	2	omicron	59.73	80.34	0.42	0.74	1.09	1.03
COVIDhub-4_week_ensemble	3	omicron	54.85	81.62	0.53	0.89	0.76	0.78
THieF_6wk-4root	3	omicron	60.72	88.81	0.32	0.81	0.83	0.85
THieF_12wk-4root	3	omicron	60.90	89.02	0.32	0.80	0.84	0.85
THieF_ensemble-mean	3	omicron	64.60	93.29	0.38	0.80	0.89	0.89
THieF_ensemble-train3	3	omicron	64.80	93.84	0.37	0.80	0.89	0.90
THieF_6wk-noTransform	3	omicron	68.19	97.82	0.33	0.75	0.94	0.94
COVIDhub-baseline	3	omicron	72.69	104.38	0.60	0.90	1.00	1.00
sarima_s1-noTransform	3	omicron	79.93	107.22	0.39	0.71	1.10	1.03
sarima_s7-noTransform	3	omicron	80.02	106.10	0.43	0.73	1.10	1.02
COVIDhub-4_week_ensemble	4	omicron	68.91	100.99	0.50	0.87	0.79	0.80
THieF_12wk-4root	4	omicron	70.92	104.68	0.29	0.79	0.81	0.83
THieF_6wk-4root	4	omicron	73.43	107.79	0.30	0.78	0.84	0.85
THieF_ensemble-mean	4	omicron	78.30	113.70	0.36	0.78	0.90	0.90
THieF_ensemble-train3	4	omicron	78.74	114.59	0.34	0.78	0.90	0.90
THieF_6wk-noTransform	4	omicron	81.60	117.92	0.31	0.73	0.93	0.93
COVIDhub-baseline	4	omicron	87.33	126.66	0.58	0.90	1.00	1.00

sarima_s1-noTransform	4	omicron	97.01	130.36	0.37	0.68	1.11	1.03
sarima_s7-noTransform	4	omicron	99.46	133.08	0.42	0.71	1.14	1.05

Table 10. Summary of model performance during the testing phase stratified by pandemic wave and horizon week for averaged states, ordered by ascending WIS

Model	hzn	wave	WIS	MAE	Cov50	Cov95	rWIS	rMAE
COVIDhub-4_week_ensemble	1	ba4_ba5	277.8	378.7	0.89	1.00	0.62	0.78
THieF_6wk-4root	1	ba4_ba5	350.4	542.9	0.60	0.99	0.78	1.11
THieF_ensemble-train3	1	ba4_ba5	399.1	685.3	0.46	1.00	0.88	1.41
THieF_ensemble-mean	1	ba4_ba5	433.6	740.3	0.49	1.00	0.96	1.52
sarima_s1-noTransform	1	ba4_ba5	439.7	531.4	0.80	1.00	0.98	1.09
COVIDhub-baseline	1	ba4_ba5	450.8	486.9	0.92	1.00	1.00	1.00
THieF_12wk-4root	1	ba4_ba5	455.6	760.8	0.42	1.00	1.01	1.56
THieF_6wk-noTransform	1	ba4_ba5	516.5	822.5	0.60	1.00	1.15	1.69
sarima_s7-noTransform	1	ba4_ba5	517.8	828.6	0.41	0.99	1.15	1.70
COVIDhub-4_week_ensemble	2	ba4_ba5	411.4	572.5	0.85	1.00	0.63	0.77
THieF_6wk-4root	2	ba4_ba5	518.8	813.0	0.61	1.00	0.79	1.10
THieF_ensemble-train3	2	ba4_ba5	578.6	966.6	0.49	1.00	0.88	1.30
THieF_12wk-4root	2	ba4_ba5	597.8	988.5	0.52	1.00	0.91	1.33
sarima_s1-noTransform	2	ba4_ba5	599.4	685.5	0.79	1.00	0.92	0.92
THieF_ensemble-mean	2	ba4_ba5	621.8	1030.3	0.50	1.00	0.95	1.39
COVIDhub-baseline	2	ba4_ba5	654.2	741.0	0.98	1.00	1.00	1.00
THieF_6wk-noTransform	2	ba4_ba5	727.8	1145.9	0.59	1.00	1.11	1.55
sarima_s7-noTransform	2	ba4_ba5	729.7	1169.6	0.53	1.00	1.11	1.58
COVIDhub-4_week_ensemble	3	ba4_ba5	563.3	824.7	0.77	1.00	0.68	0.79
THieF_6wk-4root	3	ba4_ba5	683.9	1069.1	0.59	1.00	0.83	1.03
sarima_s1-noTransform	3	ba4_ba5	725.7	824.4	0.73	1.00	0.88	0.79
THieF_ensemble-train3	3	ba4_ba5	784.0	1311.5	0.45	1.00	0.95	1.26
THieF_12wk-4root	3	ba4_ba5	797.1	1329.2	0.44	1.00	0.97	1.28
COVIDhub-baseline	3	ba4_ba5	825.0	1038.9	1.00	1.00	1.00	1.00
THieF_ensemble-mean	3	ba4_ba5	825.0	1371.3	0.50	1.00	1.00	1.32
THieF_6wk-noTransform	3	ba4_ba5	911.0	1464.4	0.59	1.00	1.10	1.41
sarima_s7-noTransform	3	ba4_ba5	953.5	1514.2	0.57	1.00	1.16	1.46
COVIDhub-4_week_ensemble	4	ba4_ba5	698.9	1034.3	0.77	1.00	0.72	0.78
THieF_6wk-4root	4	ba4_ba5	801.9	1210.3	0.64	1.00	0.82	0.92
sarima_s1-noTransform	4	ba4_ba5	826.6	917.4	0.71	1.00	0.85	0.69
THieF_ensemble-train3	4	ba4_ba5	969.8	1585.1	0.48	1.00	1.00	1.20
COVIDhub-baseline	4	ba4_ba5	972.0	1320.5	1.00	1.00	1.00	1.00
THieF_ensemble-mean	4	ba4_ba5	1010.8	1651.7	0.53	1.00	1.04	1.25
THieF_6wk-noTransform	4	ba4_ba5	1034.0	1631.7	0.61	1.00	1.06	1.24

THieF_12wk-4root	4	ba4_ba5	1035.3	1729.4	0.33	0.99	1.06	1.31
sarima_s7-noTransform	4	ba4_ba5	1169.6	1821.8	0.60	1.00	1.20	1.38
COVIDhub-4_week_ensemble	1	omicron	1067.9	1595.1	0.46	0.87	0.72	0.77
THieF_ensemble-train3	1	omicron	1455.0	2057.5	0.27	0.71	0.98	0.99
THieF_ensemble-mean	1	omicron	1462.3	2048.6	0.27	0.71	0.98	0.98
COVIDhub-baseline	1	omicron	1487.9	2083.4	0.53	0.83	1.00	1.00
THieF_6wk-noTransform	1	omicron	1576.0	2156.3	0.32	0.74	1.06	1.03
sarima_s1-noTransform	1	omicron	1588.3	2137.9	0.37	0.73	1.07	1.03
THieF_6wk-4root	1	omicron	1709.3	2355.1	0.28	0.56	1.15	1.13
sarima_s7-noTransform	1	omicron	1811.9	2287.4	0.22	0.62	1.22	1.10
THieF_12wk-4root	1	omicron	1951.3	2634.7	0.23	0.65	1.31	1.26
COVIDhub-4_week_ensemble	2	omicron	1919.4	2844.9	0.47	0.84	0.74	0.81
THieF_ensemble-mean	2	omicron	2272.2	3137.2	0.30	0.69	0.88	0.89
THieF_ensemble-train3	2	omicron	2276.3	3154.0	0.29	0.68	0.88	0.90
THieF_6wk-4root	2	omicron	2329.9	3275.3	0.22	0.69	0.90	0.93
THieF_6wk-noTransform	2	omicron	2349.2	3233.6	0.34	0.73	0.91	0.92
THieF_12wk-4root	2	omicron	2472.5	3412.2	0.22	0.63	0.96	0.97
COVIDhub-baseline	2	omicron	2579.1	3519.7	0.50	0.70	1.00	1.00
sarima_s1-noTransform	2	omicron	2777.8	3617.0	0.34	0.66	1.08	1.03
sarima_s7-noTransform	2	omicron	3146.1	3936.9	0.29	0.65	1.22	1.12
COVIDhub-4_week_ensemble	3	omicron	2757.2	4201.2	0.43	0.80	0.77	0.85
THieF_12wk-4root	3	omicron	2994.0	4176.2	0.18	0.60	0.84	0.85
THieF_6wk-4root	3	omicron	3028.8	4330.8	0.15	0.68	0.84	0.88
THieF_6wk-noTransform	3	omicron	3046.9	4276.4	0.33	0.71	0.85	0.87
THieF_ensemble-mean	3	omicron	3131.3	4321.1	0.28	0.67	0.87	0.88
THieF_ensemble-train3	3	omicron	3152.9	4350.3	0.26	0.63	0.88	0.89
COVIDhub-baseline	3	omicron	3583.0	4912.1	0.43	0.67	1.00	1.00
sarima_s1-noTransform	3	omicron	3872.9	5029.4	0.25	0.61	1.08	1.02
sarima_s7-noTransform	3	omicron	4490.1	5610.1	0.35	0.60	1.25	1.14
THieF_12wk-4root	4	omicron	3542.1	4909.0	0.12	0.58	0.80	0.81
COVIDhub-4_week_ensemble	4	omicron	3556.3	5294.6	0.37	0.79	0.80	0.87
THieF_6wk-noTransform	4	omicron	3709.9	5382.9	0.27	0.71	0.84	0.88
THieF_6wk-4root	4	omicron	3763.1	5340.8	0.10	0.60	0.85	0.88
THieF_ensemble-mean	4	omicron	3901.0	5453.5	0.23	0.62	0.88	0.90
THieF_ensemble-train3	4	omicron	3937.2	5519.3	0.17	0.60	0.89	0.91
COVIDhub-baseline	4	omicron	4439.6	6090.8	0.36	0.61	1.00	1.00

sarima_s1-noTransform	4	omicron	4810.9	6203.0	0.23	0.58	1.08	1.02
sarima_s7-noTransform	4	omicron	5781.5	7264.5	0.31	0.53	1.30	1.19

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